A Fundamental Diagram-Consistent Fluid Queue Model for Dynamic Throughput under Heavy Traffic Congestion

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Abstract: The efficient operation of transportation systems is a critical priority for policymakers, particularly given the increasingly emphasis on efficiency gains to meet growing travel demands without relying solely on capacity expansion. During peak hours at freeway bottleneck locations, a flow drop may be observed, and this drop is influenced by traffic stream attributes such as merging and diverging vehicles, resulting in significant efficiency losses. However, queue-based models, widely used for estimating delays and queue lengths, often oversimplify congestion dynamics by assuming a constant outflow rate, leading to inconsistencies when compared with FD-based observations of over-congested states. In this manuscript, we introduce a novel Fundamental Diagram-Consistent Fluid Queue (FDQ) framework for analyzing and mitigating traffic efficiency losses during heavy congestion. We extend the traditional fluid queue model by incorporating a stationary density-flow relationship observed empirically at key bottlenecks. Unlike classical queue-based models, our framework allows the flow throughput to evolve with local traffic state transitions, especially the shift from semi-congested to fully congested regimes. We start with triangular FD and show how to analytically derive FD-consistent dynamic flow throughput, as well as the associated traffic states such as the queue profile and waiting time. Such framework is then utilized to understand efficiency loss mechanisms and explore the potential for increased system efficiency through targeted inflow control. Two types of FDQ models were developed: one with flow throughput in polynomial form (FDQ-PN) and another in piecewise form (FDQ-PW). The FDQ framework is also extended to work with quadratic FD. Validation and numerical analyses were performed using datasets from Los Angeles I-405 and Phoenix I-10. The results demonstrate that the proposed framework substantially improves the accuracy of traffic

state estimation. Furthermore, the demand-supply coupled inflow control is shown to offer a more significant efficiency gain than adjusting demand or supply alone.

Keywords: Traffic state estimation, Fluid queue-based model, Fundamental diagram, Flow throughput, Outflow rate, Inflow control

1 Introduction and Motivation

The efficient operation of transportation systems is a critical priority for policymakers, especially given the increasing emphasis by the federal government on improving efficiency (USDOT 2022). In the context of transportation engineering, this priority translates to meeting growing travel demands without relying solely on expanding capacity (Downs 2005). This shift highlights the need for analytical tools that not only identify system inefficiencies but also provide actionable insights to mitigate them (Papageorgiou et al., 2003; Small et al., 2024).

In this study, we address efficiency losses in transportation systems observed during periods of severe traffic congestion. Specifically, we observe that during peak hours at freeway bottleneck locations, a flow drop often occurs, and this drop is influenced by traffic stream attributes such as merging and diverging vehicles, resulting in significant efficiency losses. In the first subsection below, we present empirical observations from the PeMS database along the U.S. I-405 corridor in Los Angeles, California. The data is collected from a detector located in a recurrently congested merging area and is not constrained by downstream traffic conditions.

We then examine the challenges of integrating this flow throughput degradation at into queue-based models and discuss the need for developing a Fundamental Diagram-Consistent Fluid Queue (FDQ) modeling approach to better capture and enhance system efficiency. Point queue-based models, widely adopted for performance evaluation, have long been effective in estimating delays and queue lengths. However, these models often oversimplify congestion dynamics by assuming a constant flow throughput (Newell 1982; Cheng et al., 2022), leading to inconsistencies with the congestion dynamics observed in empirical real-world traffic data. This limitation undermines the model's ability to capture the temporal evolution of congestion, thereby restricting its applicability in accurately representing real-world traffic dynamics. Evidence from recent CAV research (Talebpour and Mahmassani, 2016; Shi and Li, 2021) suggests that cooperative merging and smoother inflows can alleviate turbulence-induced capacity drop, underscoring the importance of FD-governed models that integrate demand-supply interactions.

To address this gap, the proposed FDQ could analyze and quantify efficiency loss due to flow throughput degradation during congestion period. This framework also enables the exploration of opportunities to improve system efficiency through targeted inflow adjustments, including strategies such as reservation systems and connected and CAV-enabled cooperative adjustments. By capturing the dynamic interplay between inflow rates (i.e., demand), flow throughput (i.e., supply), and congestion propagation, the FDQ framework provides a more detailed understanding of the mechanisms driving efficiency losses and identifies actionable strategies to enhance system efficiency through inflow control. In this study, we focus on the impact of traffic efficiency, including congestion duration mitigation, maximum queue length reduction, delay minimization, and overall network performance enhancement.

1.1 Empirical motivation for efficiency loss in the over-congested regime

We motivate this study using PeMS data from the I-405 corridor (May 12, 2017). The bottleneck at milepost 13.74 near Brookhurst Street exhibits recurrent peak-hour congestion without downstream constraints. Detector and downstream details appear in Section 5.1.

Fig. 1(a) illustrates that in the over-congested regime (red dots), traffic flow behaves differently from typical free-flow conditions (blue dots), with density increasing as flow decreases. Further, in Fig. 1(b) we show the temporal evolution of flow (solid line) and density (scattered dots). It can be observed that during the over-congestion period $[t_0, t_3]$ (12:20 to 20:20), traffic

flow follows a "breakdown-then-recovery" pattern, while density exhibits a "rise-then-fall" trend. These opposing dynamics indicate that traffic conditions are highly variable during congestion, with flow release rates fluctuating rather than remaining constant. This highlights significant efficiency loss under unmanaged excessive demand. Such flow breakdown has been consistently observed in prior studies (Cassidy and Bertini, 1999; Chung et al., 2007) across both individual links and broader networks during periods of heavy congestion. In Section 5.1, we have demonstrated in detail that there is no congestion downstream of the bottleneck area selected in this paper.

Building on the preceding analysis, we fit the observed flow throughput using both a constant and a dynamic throughput function. As shown in Fig. 1(c), the dynamic fitting (red curve) captures temporal variations in congested flow more accurately than the constant throughput model (green line), which fails to reflect such fluctuations. Over 81 days of data, the dynamic throughput approach achieves a significantly lower average RMSE (105.85 vs. 167.15), as shown in Fig. 1(d), demonstrating its superior performance in modeling congested traffic states. Based on the empirical observations above, Fig. 1(e) illustrates the concept of FDQ modeling framework and its application in quantifying efficiency loss and achieving system efficiency gains. Dynamic coupling of inflow $\lambda(t)$ and outflow $\mu(t)$ enables a more realistic demand-supply optimization framework.

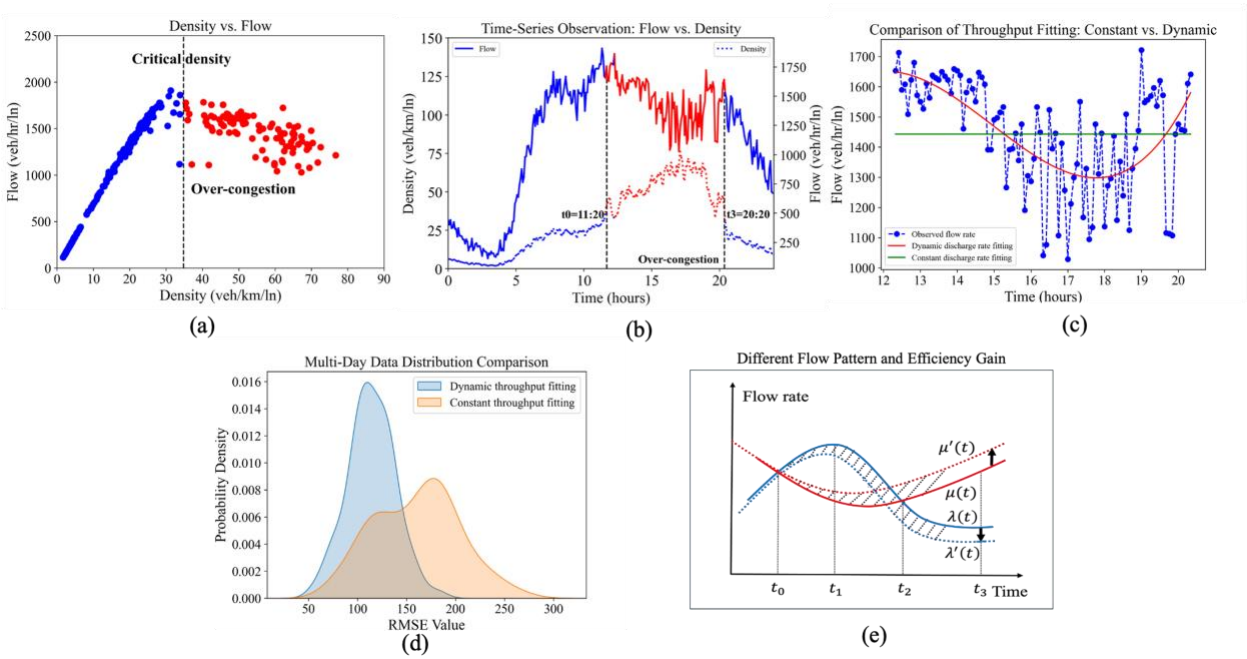


Figure 1 Empirical motivation. (a) flow-density FD; (b) time-dependent flow and density profile; (c) flow throughput fitting during congestion period; (d) fitting error distribution; (e) potentials for efficiency gain.

1.2 Research approach and theoretical contribution

One approach to addressing these empirical observations is to advocate for the development of new FD-consistent fluid queue models that incorporate the modeling of dynamic flow throughput. For the sake of a coherent presentation, we adopt the triangular fundamental diagram in the main body of this manuscript and develop two types of FDQ models: one with polynomial

throughput flow (FDQ-PN) and another in piecewise form (FDQ-PW). However, the modeling framework has also been extended to work with the quadratic fundamental diagram, and this is discussed in Section 4.1.4. In the main body of this manuscript, we use the Los Angeles dataset for the case study to validate the proposed model and demonstrate its effectiveness.

From a methodological development standpoint, we hope this research opens new avenues for cross-resolution modeling. This effectively uncovers the intrinsic relationship between queue dynamics and inflow control. This theoretical exploration is crucial for understanding the factors contributing to efficiency loss and identifying conditions for achieving efficiency gain through targeted inflow control. From a practical application standpoint, we hope that the developed modeling framework will enhance overall traffic efficiency and provide a strong foundation for multi-level transportation system management and operations (TSMO) strategies.

The remainder of this paper is organized as follows: Section 2 reviews related work on over-congestion estimation, and dynamic flow evolution estimation. Section 3 provides some preliminary concepts and discusses how this study diverges from the prior modeling approach. Section 4 introduces the proposed methodology, beginning with the derivation of the FD-consistent flow throughput, followed by the development of the FDQ model, the derivation of essential parameters, and concluding with the modeling of efficiency gains. Section 5 presents the case study and analyzes the performance of the model. Section 6 presents the discussion of the model. Finally, Section 7 concludes this research and suggests directions for future research.

2 Literature Review

The fundamental diagram defines the relationships among speed, flow, and density, forming the core of traffic flow theory. Early models, starting with Greenshields et al. (1935), used a linear speed-density relation. Later models, such as Greenberg's logarithmic, Underwood's exponential (1961), and Drew's curve-based formulations (1964), offered single-regime representations across all densities. While elegant, these models often trade simplicity for accuracy, performing well under certain conditions but less so in others.

Newell (1993a, 1993b, 1993c) proposed a simplified triangular FD that captures the relationship among traffic flow, speed, and density, significantly reducing the complexity of kinematic wave theory. In this model, queuing begins only after the peak point, which represents capacity and marks the traffic transitions from free-flow to over-congested conditions. Flow near the front of the queue is termed "forced flow," while flow at the back is "normal flow," reflecting distinct traffic states across the link. The triangular FD simplifies traffic interactions and facilitates the analysis of congestion behavior. Variants of the FD have been developed for different contexts, such as urban expressways (Van Aerde, 1995), heavy congestion (Cheng et al., 2021), and microscopic car-following behavior (Ni et al., 2016). While the FD effectively describes steady-state relationships, its ability to represent dynamic traffic evolution is limited. Integrating it with a queueing model enables more realistic modeling of congestion buildup and dissipation. The following subsections review research on over-congestion estimation and queue evolution to address these dynamics.

2.1 Literature review on over-congestion estimation

Numerous studies (Cassidy and Bertini, 1999; Zhang and Levinson, 2004; Chung et al., 2007; Chen and Ahn, 2018) have identified the "capacity drop" or "flow breakdown" phenomenon at bottlenecks, where roadway capacity decreases once queues form. Based on empirical observations at freeway merge bottlenecks, Laval et al. (2007) found that the formation of

upstream queues leads to a reduced discharge rate, a phenomenon known as "capacity drop", which is accompanied by increased lane-changing activity and can be mitigated through ramp metering. The capacity drops issue, initially analyzed from a dynamic equilibrium perspective by Vickrey (1969) and later expanded by Arnott et al. (1990), demonstrates how drivers balance time delays with schedule constraints to optimize departure times, though these models do not address over-congestion. Marked by low speed and high density that led to suboptimal traffic flow, accurate modeling of over-congestion is a critical factor in congestion mitigation. Several studies have examined over-congested traffic conditions, often at the link level and in conjunction with the FD. Small (2003) developed a model for central business districts using empirical speed-density relationships to characterize over-congestion, and later, Small and Verhoef (2007) noted that similar flow levels can emerge from both congested and over-congested states. Lo and Szeto (2005), using shockwave analysis, argued that backward-bending functions are inadequate for modeling over-congestion and road pricing. Over-congestion is not limited to highway bottlenecks (Anderson and Davis, 2020) but also frequently occurs at merge and diverge points.

The over-congestion phenomenon, where traffic density increases as throughput reduces, is well understood at the link level and has also been observed macroscopically in traffic networks by Frascaria et al. (2020). While the macroscopic fundamental diagram (MFD) describes over-congestion, it does not explain the dynamic processes leading to it or represent queues during congestion (Geroliminis and Daganzo, 2008). In over-congestion analysis, particularly when using the FD, it is essential to focus on the dynamic characteristics of traffic flow during congestion to better capture flow patterns and dynamic queuing behaviors.

Dynamic modeling of congestion provides a comprehensive approach to estimating traffic congestion by incorporating spatial dynamics. Building on Vickrey's pioneering continuous-time point-queue models, which describe the dynamic interplay between arrivals and departures, various subsequent models (Han et al., 2013; Ban et al., 2012; Li et al., 2014) have examined queuing mechanisms in traffic systems with single bottlenecks. These models provide insights into how traffic congestion propagates and accumulates during peak hours. Gao and Levinson (2024) introduce a spatial queuing model for a single bottleneck during morning peak hours using a multi-stage approach, though they do not account for differences between recovery and breakdown phases. Other approaches include N-curves and T-curves (Cassidy and Bertini, 1999) and the use of cutoff speed (Wu et al., 2021; Zhou et al., 2022) to define congestion duration and queued demand.

2.2 Literature review on dynamic flow evolution estimation

When traffic demand exceeds supply in the short term and within certain spatial constraints, congestion occurs in the traffic system. As the arrival flow rate decreases, traffic density is reduced, and queued vehicles gradually pass through the congested bottleneck until free-flow conditions are restored. The process from the onset of congestion to its resolution represents a dynamic spatiotemporal evolution involving both the formation and dissipation of queues. Methods for estimating traffic congestion queues are generally divided into three categories: analytical methods, simulation methods, and hybrid approaches. Traditional volume-delay functions, such as the BPR function, primarily examine the relationship between average travel time and flow on road segments from a macroscopic perspective, focusing on the overall traffic flow characteristics. However, these functions fail to capture the dynamic queuing process that occurs under congested conditions (Bliemer et al., 2014). Therefore, when modeling and analyzing bottleneck congestion in greater depth, simulating or analytically modeling the formation and dissipation of queues using

various approaches is essential for a more accurate understanding of congestion bottleneck evolution.

The Vickrey (1969) bottleneck model is a foundational framework for analyzing peak-period congestion due to its simplicity and analytical tractability. Building on this foundation, subsequent studies have extended the model to address dynamic traffic assignment (e.g., Li et al., 2000; Kuwahara and Akamatsu, 1997; Huang and Lam, 2002) and compared it with other approaches such as delay functions and cell transmission models (Nie and Zhang, 2005). Cassidy and Rudjanakanoknad (2005) observed that discharge flow reductions often begin at shoulder lanes near merge points, driven by lane-changing behavior. Enhancements to the model include Ban et al. (2012) continuous-time differential complementarity formulation and Jin's (2015) unified framework. Other extensions account for heterogeneous traffic (Lindsey, 2004), elastic demand underpricing (Yang and Huang, 1997), and queue-capacity interactions (Chen et al., 2019). Yperman (2007) developed a dynamic link transmission model consistent with kinematic wave theory, while stochastic queuing models (e.g., Berman et al., 1987) provide probabilistic representations, though their applicability is limited under uncertain arrival patterns.

Vickrey's bottleneck model, which assumes piecewise constant arrival and dissipation rates, falls short in capturing the nonlinear dynamics of real-world traffic. To address this, Newell (1968a, 1968b, 1968c, 1982) introduced a fluid queuing model with a quadratic arrival rate and constant dissipation rate, laying the foundation for analytical bottleneck modeling. As Hall (1991) noted, deterministic fluid approximations are essential for describing non-stationary queuing systems dominated by predictable fluctuations. While Newell's model effectively captures mild congestion, it lacks the ability to represent severe congestion dynamics. To address limitations in traditional models, Cheng et al. (2022) proposed a cubic polynomial arrival rate model capable of simulating oversaturated traffic conditions and estimating key performance metrics such as time-dependent queue length, delay, and total travel time. Subsequent extensions include Tang et al. (2021), who adapted the model for bus-only lanes, and Bliemer et al. (2014), who incorporated residual queues and spillback effects by considering capacity and storage constraints. Ramezani and Geroliminis (2015) modeled queue evolution at urban intersections using probe vehicle data, while Lu et al. (2023) applied physics-informed neural networks for joint estimation of traffic states and queuing profiles. Most recently, Doll et al. (2024) extended the framework to signalized intersections with time-dependent stochastic arrivals and departures. However, these models often assume time-dependent polynomial arrival rates while overlooking polynomial departure patterns and lack integration with the FD, limiting their ability to comprehensively estimate traffic queuing systems across multiple parameters.

3 Preliminaries

In this section, we present some preliminary concepts from the literature that are related to our work and discuss how this study diverges from them. For ease of reference, the mathematical notations used throughout this paper are summarized in Appendix A.

Consider a dynamic system with a bottleneck restricting the passing of moving vehicles. Fig. 2 illustrates the dynamic queue evaluation process: Fig. 2(a) illustrates the monitoring setup at a freeway bottleneck, including loop detectors, general vehicle population, probe vehicles and the bottleneck location. Fig. 2(b) illustrate the vehicle trajectories in the space-time plane across the queue extent along a single bottleneck. Fig. 2(c) illustrates the classic constant throughput queueing model proposed by Newell (1982). The t_0 and t_3 represent the start and end of the congestion period, respectively; t_1 denotes the time with maximum arrival rate, and t_2 indicates

the time of maximum queue length. The arrival rate function is $\lambda'(t)$ at the bottleneck, and the red curve constant throughput is μ. The blue curve $\lambda'(t)$ and red horizontal line μ intersect twice, at time t_0 and t_2, respectively. Prior to time t_0, the $\lambda'(t) < \mu$, indicating non-congested traffic conditions. Between t_0 and t_2, $\lambda'(t) > \mu$ so some vehicles will have to wait in the queue. The corresponding cumulative arrival $A'(t)$ and cumulative departure curves $D'(t)$ are shown in Fig. 2(e). The cumulative departure curve increases linearly due to the constant flow throughput.

According to the empirical observation in Fig. 1, we model the throughout as a dynamic quantity to more closely capture real-world behavior, as shown in Fig. 2(d). The corresponding cumulative arrival $A(t)$ and departure curves $D(t)$ are shown in Fig. 2(f).

The dynamic throughput function $\mu(t)$ from the field observation has the following properties: (1) The dynamic throughput function $\mu(t)$ remains continuous during the congestion period $[t_0, t_3]$ and stays below the bottleneck capacity Cap except at the onset and clearance of congestion, when $\mu(t_0) = \mu(t_3) = Cap$. At the intermediate time t_2, $\mu(t)$ and $\lambda(t)$ intersect, reflecting a momentary balance between arrival demand and discharge flow. These features are consistent with empirical observations and characterize the typical temporal evolution of bottleneck throughput. (2) By t_3, all queued vehicles are fully dissipated, i.e., $\int_{t_0}^{t_2}[\lambda(t) - \mu(t)]dt = \int_{t_2}^{t_3}[\mu(t) - \lambda(t)]dt$. Since the queue dynamics are governed by the net-flow $\pi(t) = \lambda(t) - \mu(t)$, any change in the arrival pattern inevitably influences the discharge flow μ(t), and vice versa. This interdependence underlies the formation and dissipation of congestion at the bottleneck.

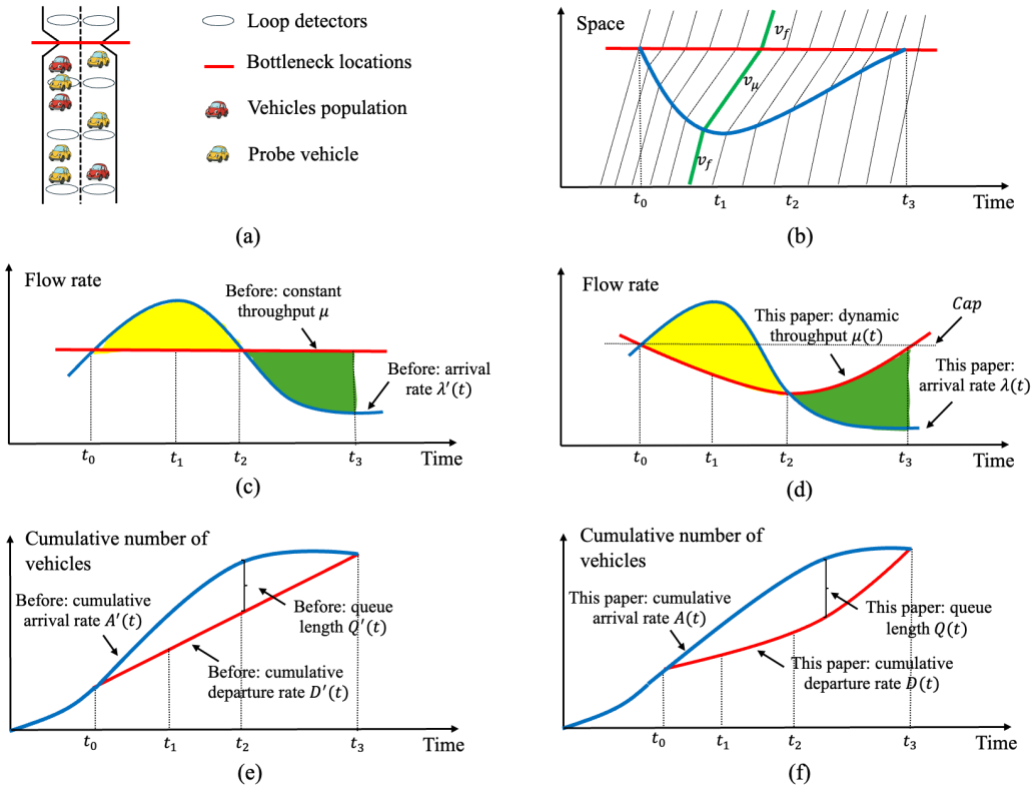


Figure 2 Illustration of the constant-throughput and dynamic-throughput queue models.

4 Theoretical Framework

4.1 Fundamental diagram-based throughput formulations for the FDQ model

We begin by illustrating the proposed method using the triangular FD and subsequently extend it to the quadratic FD. In the triangular FD below, as shown in Fig. 3(a), we denote the backward wave speed as w_b, the critical density as k_c, and the jam density as k_j. We denote the bottleneck capacity as Cap, and $Cap = k_c \cdot v_f = w_b \cdot (k_j - k_c)$. The triangular FD can be illustrated as Eq. (1), in which we use $k(t)$ to denote the time-dependent density and $q(t)$ denotes the time-dependent flow rate. In the congestion regime the observed traffic flow $q(t)$ equals the flow throughput $\mu(t)$, i.e., $q(t) = \mu(t)$ when $k(t) > k_c$. The quadratic FD is shown in Fig. 3(b).

$$q(t) = \begin{cases} v_f \cdot k(t), & for\ k(t) \le k_c \\ w_b \cdot (k_j - k(t)), & for\ k(t) > k_c \end{cases} \quad (1)$$

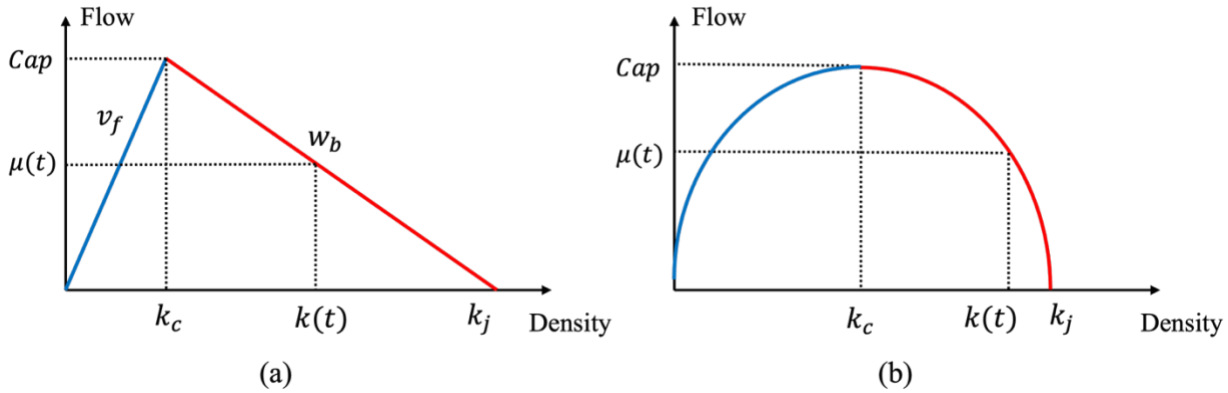


Figure 3 Illustration of the triangular and quadratic fundamental diagram.

4.1.1 FDQ formulation under the triangular fundamental diagram

Under the triangular FD, the flow throughput under congestion period is governed by the backward wave speed and the excess density above the critical density. According to Eq. (1), in the congested regime $k(t) > k_c$, the FD implies the classical relationship:

$$q(t) = \mu(t) = w_b \cdot [k_j - k(t)] \quad (2)$$

and at critical density $k = k_c$, the capacity can be represented as:

$$Cap = w_b \cdot [k_j - k_c] \quad (3)$$

Subtracting Eq. (3) from Eq. (2), we could derive the following:

$$\mu(t) = Cap - w_b \cdot [k(t) - k_c] \quad (4)$$

This expresses the well-known FD result that flow throughput decreases as density exceeds the critical value during congestion period.

To maintain consistency with the fundamental diagram, the dynamic throughput must satisfy the following equation during congestion period:

$$0 \leq \mu(t) \leq Cap, t\epsilon[t_0, t_3] \quad (5)$$

This constraint ensures that the FDQ discharge never exceeds the FD-implied bottleneck capacity.

4.1.2 Practical mapping between excess density and queue length

Assumption 1: Linear mapping between queue length and excess density.

We assume a linear mapping between queue length and excess density during the congested regime, expressed as:

$$Q(t) = \varphi \cdot [k(t) - k_c] \quad (6)$$

where φ is the conversion factor linking density changes in the fundamental diagram to the dynamic queue representation used in the FDQ model.

This assumption provides an analytically tractable bridge between macroscopic FD-based density evolution and mesoscopic queue representation. Queue length measures temporal accumulation of delayed vehicles, while density describes spatial concentration. The linear form does not imply that the physical relationship is inherently linear; instead, it approximates the proportionality implied by classical kinematic-wave theory for backward-moving congestion fronts. The coefficient φ is calibrated empirically to minimize the discrepancy between observed density evolution and the FDQ queue profile.

This linearity does not assume that traffic physics are inherently linear. Rather, it reflects a projection of the triangular FD onto a mesoscopic queue representation, allowing the FD-based density evolution to be expressed in terms of queue length at a point bottleneck. This approximation enables an FD-consistent queueing model without spatial discretization, while remaining consistent with classical kinematic wave theory.

A linear mapping between queue length and excess density has long been adopted in classical traffic flow theory. Under the triangular fundamental diagram, Newell (1993) showed that when a backward kinematic wave propagates upstream at a constant wave speed, the spatial extent of the queue is proportional to the excess density above the critical value. Similarly, Daganzo's Cell Transmission Model (CTM) (Daganzo, 1994, 1995a, 1995b) establishes that in a homogeneous link, the number of vehicles in congestion is directly governed by the difference between the jam density and the prevailing density at the bottleneck. These formulations both imply a linear relationship between queue accumulation and variations in density within the over-congested regime.

4.1.3 From density-based representation to throughput reduction

According to Eq. (6), given that:

$$k(t) - k_c = \frac{Q(t)}{\varphi} \quad (7)$$

Substituting this expression into the flow throughput relation yields in Eq. (4):

$$\mu(t) = Cap - w_b \cdot \frac{Q(t)}{\varphi} \tag{8}$$

Defining the queue impact coefficient as:

$$\theta = w_b \cdot \frac{1}{\varphi} \tag{9}$$

Assumption 2: Linear throughput reduction with respect to queue length.

We assume that the congested flow throughput decreases linearly with queue length,

$$\mu(t) = Cap - \theta \cdot Q(t) \tag{10}$$

This assumption enforces consistency with the triangular FD, whose congested branch exhibits a linear decline in flow as density exceeds the critical value. Together with Assumption 1, this ensures that $\mu(t)$ remains FD-governed, capacity-bounded, and monotonically decreasing under increasing congestion. The parameter θ reflects the sensitivity of throughput flow to queue growth and is directly calibrated from empirical bottleneck data.

4.1.4 Extension to a quadratic fundamental diagram

While Section 4.1.1-4.1.4 focused on the triangular FD, the FDQ formulation is not limited to piecewise-linear flow-density relationships. Fig. 3(b) illustrates a quadratic FD, where flow increases with density in uncongested conditions and decreases parabolically once the critical density is exceeded.

The Fig. 3(b) visually demonstrates the relationship between density and flow, illustrating a typical traffic flow pattern: flow increases with density in low-density conditions, but congestion causes a decrease in flow beyond the critical density. It demonstrate that the quadratic FD is consistent with the fluid-based queue model, indicating that under the quadratic FD, the flow throughput $\mu(t)$ decreases quadratically with the increase of the queue length $Q(t)$, with the reduction magnitude related to the road capacity Cap and the jam density k_j.

In addition to its mathematical form, the quadratic FD also carries clear physical meaning that aligns well with field-observed congestion patterns. Unlike the triangular FD, which imposes an abrupt slope change at the critical density, the quadratic FD provides a smooth transition from uncongested to over-congested conditions. This curvature reflects empirical phenomena such as gradual breakdown, progressive turbulence near merging points, and nonlinear increases in lane-changing interference. These features often appear in recurrent bottlenecks where capacity does not drop instantaneously but degrades smoothly over time. Importantly, this FD remains fully consistent with the FDQ modeling principle, because the FDQ framework only requires that the throughput flow be governed by an FD-based relationship between flow and density. The quadratic FD satisfies this requirement by ensuring that throughput $\mu(t)$ is bounded by capacity and decreases monotonically with excess density. When projected through the queue length and density mapping, this results in a quadratic throughput-queue length relationship that maintains FD-consistent queue evolution.

Under the quadratic FD, the flow throughput $\mu(t)$ in the congested regime $k(t) > k_c$ follows:

$$\mu(t) = Cap \cdot \left[1 - \left(\frac{k(t) - k_c}{k_j - k_c}\right)^2\right] \tag{11}$$

Assuming $k_j \gg k_c$ (jam density is much larger than critical density), we approximate $(k_j - k_c) \approx k_j$. And we could derive the following:

$$\mu(t) = Cap \cdot \left[1 - \left(\frac{Q(t)}{k_j}\right)^2\right] \tag{12}$$

Expend and rewrite Eq. (12), we could derive the following:

$$\mu(t) = Cap - \frac{Cap}{k_j^2} \cdot [Q(t)]^2 \tag{13}$$

Under the quadratic FD, the flow throughput $\mu(t)$ decreases quadratically with the increase of the queue length $Q(t)$, with the reduction magnitude determined by Cap and k_j. Therefore, the quadratic FD example further demonstrates that the FDQ modeling principle is general and not tied to a particular FD shape.

4.2 Dynamic-throughput queueing model formulation

With the arrival rate derived above, in this section we present the development of FDQ model and the derivation of essential parameters for traffic state estimation. We will start from FDQ models in polynomial form (FDQ-PN) and then move onto a special simplified scenario with queue models in piecewise form (FDQ-PW) in Appendix C.

In this section, we adopt a polynomial form to represent the arrival and departure rates. Extending Newell's work (Newell 1982), we introduce a generalized model that replaces the constant departure rate with a cubic profile, providing a more accurate representation of traffic system dynamics. Fig. 4 is used for illustration purpose, followed by mathematical derivations.

In Fig. 4(a), the blue and red curves represent the arrival rate $\lambda(t)$ and flow throughput rate $\mu(t)$, respectively. Their mathematical expressions will be derived in the subsections below. These two curves intersect twice, at t_0 and t_2, respectively. Correspondingly, Fig. 4(b) shows the cumulative curves at the bottleneck, where the blue curve represents the cumulative arrival rate $A(t)$, and the red curve represents the cumulative departure rate $D(t)$. The queue profile is depicted in Fig. 4(c), with the blue curve illustrating the process of queue formation and dissipation over time. Consistent with the above definition of time, the length of queue is zero at t_0 and t_3, and reach highest at time t_2. Finally, Fig. 4(d) represents the net flow $\pi(t) = \lambda(t) - \mu(t)$. We use the term "net flow" to denote the difference between inflow and outflow at the bottleneck, consistent with queueing representations.

Although the FDQ model introduces four time-based parameters to describe the onset, peak formation, equilibrium, and dissipation of congestion, these parameters should not be interpreted

as quantities that are identifiable only ex post. Instead, they represent transition points in the demand-supply imbalance, and their values can originate from (i) empirical detection, (ii) short-term prediction models, or (iii) controller-specified operational planning.

From an empirical standpoint, t_0 and t_3 can be obtained in real time using standard tools such as cutoff-speed detection, cumulative count curves, or detector-based congestion identification modules, all of which are widely implemented in modern TMCs. For predictive applications, these timestamps can be supplied by short-term forecasting models (e.g., for predicted demand surges) or by Active Traffic and Demand Management (ATDM) logic in which controllers determine the desired onset or termination of congestion under alternative strategies. Thus, the parameters are not restricted to ex-post calibration; they serve as structural descriptors that can be informed by either observed or predicted conditions.

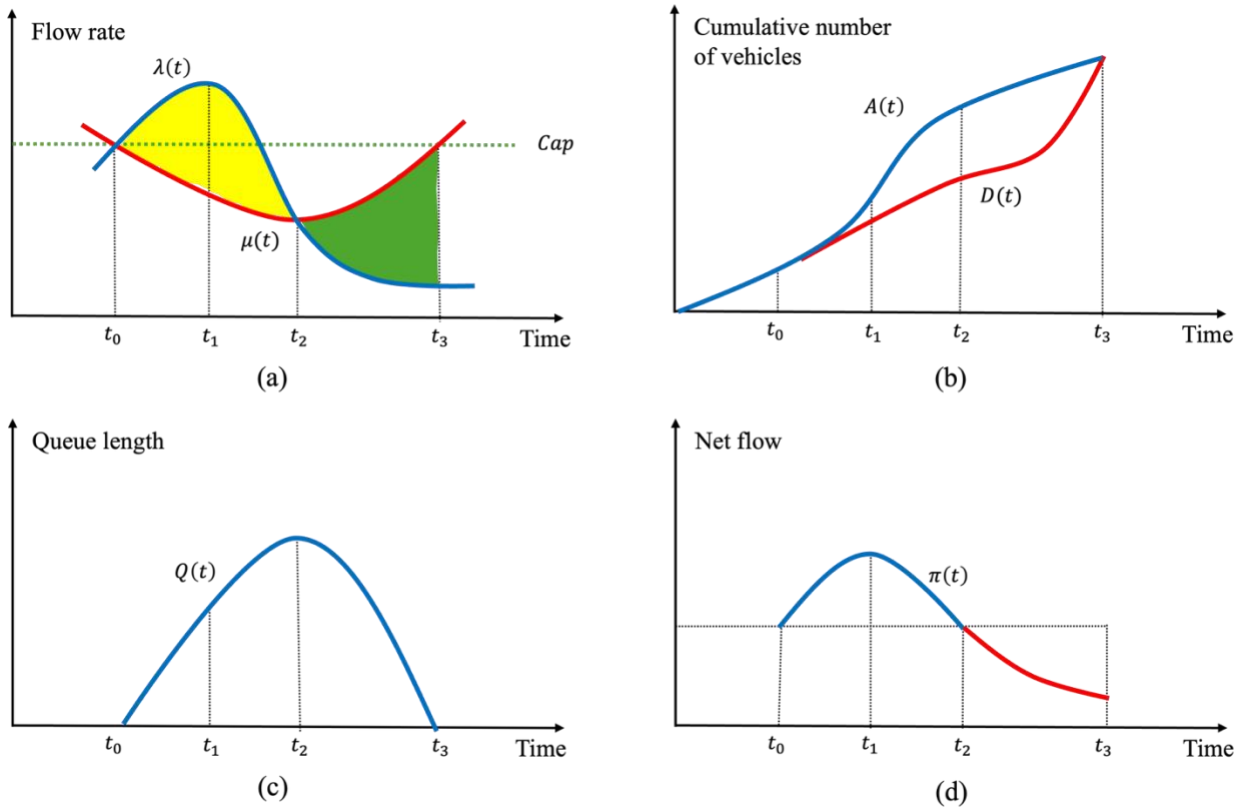


Figure 4 Illustration of FDQ-PN with polynomial queue models: (a) arrival and departure rate; (b) cumulative arrival and departure rate; (c) queue length profile; (d) net flow profile.

Proposition 1: When expressing the time-dependent arrival rate $\lambda(t)$ and the flow throughput $\mu(t)$ as cubic polynomial functions:

a) Their cubic terms have the same coefficients. Furthermore,

b) The net flow $\pi(t)$ is a quadratic function, with a degree one less than both $\lambda(t)$ and $\mu(t)$.

Consequently,

c) The queue length $Q(t)$ will also be a cubic function, having the same degree as both $\lambda(t)$ and $\mu(t)$.

Proof: Let the arrival rate $\lambda(t)$ and the flow throughput $\mu(t)$ be cubic polynomial functions with leading coefficients a_1 and a_2, respectively, along with other parameters of b, c, and d, as shown below:

$$\lambda(t) = a_1 \cdot t^3 + b_1 \cdot t^2 + c_1 \cdot t + d_1 \tag{14}$$

$$\mu(t) = a_2 \cdot t^3 + b_2 \cdot t^2 + c_2 \cdot t + d_2 \tag{15}$$

Let's first examine the net flow profile $\pi(t)$ depicted in Fig. 4(d). During a specific episode spanning the interval $[t_0, t_3]$, this profile intersects the horizontal line $\pi = 0$ on two distinct occasions, indicating that the equation $\pi(t) = 0$ possesses two solutions. Based on such property, we can express the net flow function $\pi(t)$ in its factored form using the parameters γ, t_0 and t_2, where γ represents the shape parameter.

$$\pi(t) = \gamma(t - t_0)(t_2 - t) \tag{16}$$

The shape parameter γ determines the curvature of the net-flow profile and therefore controls how sharply the queue builds up and dissipates within the congestion window. A larger γ results in a steeper net inflow-outflow imbalance, producing a taller but narrower queue peak, whereas a smaller γ generates a flatter and more spread-out congestion pattern. Because inflow, outflow, queue length, cumulative flows, waiting time, and delay are all analytically derived from the net-flow function through integration, the accuracy of these quantities depends directly on how accurately γ is calibrated from the observed data.

Next, we derive the time-dependent queue length $Q(t)$, which equals to the integral of the net flow from t_0 to t. In other words, we have $Q(t) = \int_{t_0}^{t} \pi(\tau)d\tau = \int_{t_0}^{t} \gamma(\tau - t_0)(t_2 - \tau)d\tau$.This looks a bit intermediating.

We will apply the substitution method, letting $u = t - t_0$, which gives us $du = dt$, and this equation becomes $Q(t) = \int_{t_0}^{t} \gamma(\tau - t_0)(t_2 - \tau)d\tau = \int_{0}^{u} \gamma u[t_2 - (u + t_0)]\, du = \gamma(t_2 - t_0)\frac{1}{2}u^2 - \gamma\frac{1}{3}u^3$. Then, bringing $u = t - t_0$ back into the above equation, and with some reorganization, we will have Eq. (17) below. For detailed derivation, please refer to Appendix B.1.

$$Q(t) = \gamma(t - t_0)^2\left[\frac{t_2 - t_0}{2} - \frac{t - t_0}{3}\right] \tag{17}$$

In Fig. 4(b), we notice that the queue fully dissipates at time t_3, i.e., $Q(t_3) = 0$. This observation allows us to derive the relationship between critical time points:

With $Q(t_3) = \gamma(t_3 - t_0)^2\left[\frac{t_2 - t_0}{2} - \frac{t_3 - t_0}{3}\right] = 0$, and obviously $t_3 - t_0 \neq 0$, we will have $\frac{t_2 - t_0}{2} - \frac{t_3 - t_0}{3} = 0$. Then, with some reorganization we will have the following relationship.

$$t_2 = \frac{1}{3}t_0 + \frac{2}{3}t_3 \tag{18}$$
$$t_3 = \frac{3}{2}(t_2 - t_0) + t_0$$

Substitute $t_2 = \frac{1}{3}t_0 + \frac{2}{3}t_3$ into the expression for $Q(t)$ in Eq. (17), we could derive:

$$Q(t) = \gamma(t - t_0)^2\left[\frac{t_2 - t_0}{2} - \frac{t - t_0}{3}\right] = \gamma(t - t_0)^2\left[\frac{\frac{1}{3}t_0 + \frac{2}{3}t_3 - t_0}{2} - \frac{t - t_0}{3}\right] \tag{19}$$
$$= \gamma(t - t_0)^2\left[\frac{t_3 - t_0}{3} - \frac{t - t_0}{3}\right] = \frac{\gamma}{3}(t - t_0)^2(t_3 - t)$$

We will move onto the derivation of the cubic form flow throughput $\mu(t)$ and the arrival rate $\lambda(t)$. For $\mu(t)$, we have $\mu(t) = Cap - \theta \cdot Q(t)$. If we substitute the $Q(t)$ with the expression from Eq. (10), we will get:

$$\mu(t) = Cap - \theta \cdot \frac{\gamma}{3}(t - t_0)^2(t_3 - t) \tag{20}$$

Additionally, with the definition of network flow, we have $\pi(t) = \lambda(t) - \mu(t)$. In other words, we can compute $\lambda(t)$ with $\mu(t) + \pi(t)$. The complete equation is shown below.

$$\lambda(t) = \gamma(t - t_0)\left(\frac{1}{3}t_0 + \frac{2}{3}t_3 - t\right) + Cap - \theta \cdot \frac{\gamma}{3}(t - t_0)^2(t_3 - t) \tag{21}$$

By observing Eqs. (20)-(21), if we were to rearrange them into the polynomial form of Eqs. (14)-(15), we can find that their cubic terms are $a_1 = \theta \cdot \frac{\gamma}{3}$ as well as $a_2 = \theta \cdot \frac{\gamma}{3}$. In other words, we have $a_1 = a_2$, which completes the proof of Proposition 1-a. Additionally, by observing Eq. (10), the net flow $\pi(t)$ is obviously in a quadratic function, with a degree one less than both $\lambda(t)$ and $\mu(t)$ and the proof of Proposition 1-b is complete. By observing Eq. (13), the queue length $Q(t)$ is a cubic function with a cubic-term coefficient of $-\frac{\gamma}{3}$ and the proof of Proposition 1-c is complete. Combined, the proof of Proposition 1 is complete.

Until this moment, we have $\lambda(t)$, $\mu(t)$, $\pi(t)$ and $Q(t)$ analytically derived. Next, we will move onto deriving the other parameters for traffic state estimation in the FDQ modeling framework, including the cumulative departure flow rate $D(t)$, cumulative arrival flow rate $A(t)$, time-dependent waiting time $tt(t)$, and the total delay W.

For the cumulative departure flow rate $D(t)$: since it equals to the integral of the departure flow rate in time period, we can obtain $D(t) = \int_{t_0}^{t}\mu(\tau)d\tau = \int_{t_0}^{t}\left[Cap - \theta \cdot \frac{\gamma}{3}(\tau - t_0)^2(t_3 - \tau)\right]d\tau$. Again, this looks a bit intermediating, and similarly, we will utilize the substitution method and let $u = \tau - t_0$. In the end, we will obtain the following function. For detailed derivation, please refer to Appendix B.2.

$$D(t) = \int_{t_0}^{t} \mu(\tau)d\tau = \int_{t_0}^{t} \left[Cap - \theta \cdot \frac{\gamma}{3}(\tau - t_0)^2(t_3 - \tau)\right] d\tau = \tag{22}$$
$$= Cap \cdot (t - t_0) - \theta \cdot \frac{\gamma}{3} \int_0^{t-t_0} u^2[(t_3 - t_0) - u]du$$
$$= Cap \cdot (t - t_0) - \theta \cdot \frac{\gamma}{3} \cdot (t - t_0)^3 \cdot \left(\frac{t_3 - t_0}{3} - \frac{t - t_0}{4}\right)$$

For the cumulative arrival flow rate $A(t)$: since it equals to the integral of the arrival flow rate, we can obtain the following function. For detailed derivation, please refer to Appendix B.3.

$$A(t) = \int_{t_0}^{t} \lambda(\tau)d\tau \tag{23}$$
$$= \int_{t_0}^{t} [\gamma(\tau - t_0)\left(\frac{1}{3}t_0 + \frac{2}{3}t_3 - \tau\right) + Cap - \theta \cdot \frac{\gamma}{3}(\tau - t_0)^2(t_3 - \tau)]d\tau$$
$$= \gamma\left(\frac{1}{3}t_0 + \frac{2}{3}t_3\right) \cdot \frac{(t - t_0)^2}{2} - \gamma(t - t_0)^2\left(\frac{t - t_0}{3} - \frac{t_0}{2}\right) + Cap \cdot (t - t_0) - \theta \cdot \frac{\gamma}{3} \cdot (t - t_0)^3 \cdot \left(\frac{t_3 - t_0}{3} - \frac{t - t_0}{4}\right)$$

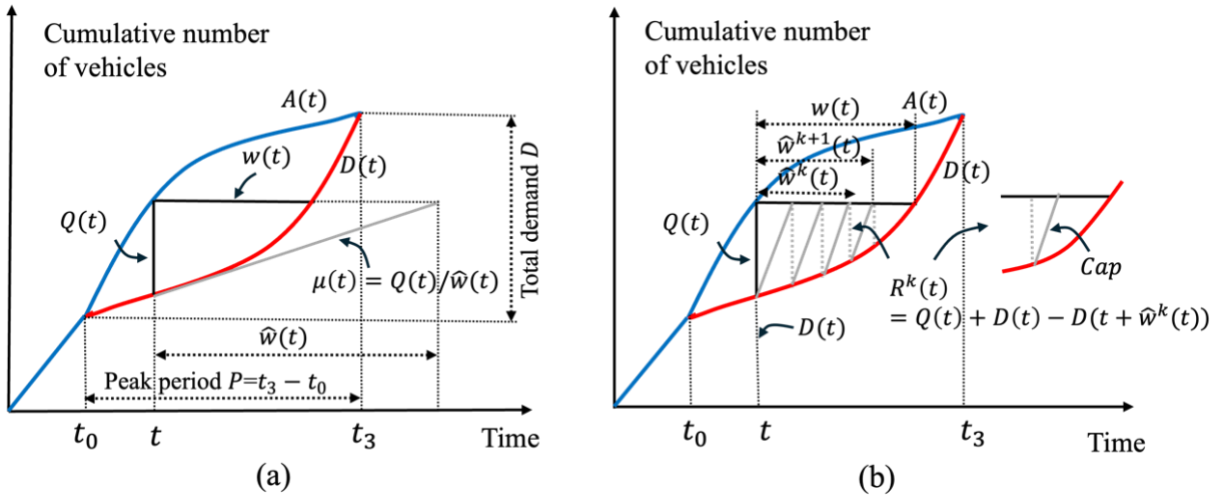


Figure 5 Time-dependent delay derivations: (a) ratio method; (b) iterative method.

Next, we derive the time-dependent waiting time $w(t)$. As illustrated in Fig. 5(a), which is a magnified version of Fig. 4(c), the blue line represents cumulative arrival rate $A(t)$ and the red line represents cumulative departure rate $D(t)$. The ratio method, proposed by Newell (1982) and later utilized in Cheng et al. (2022), computes $\frac{Q(t)}{\mu(t)}$ to determine $w(t)$ directly. However, this approach can be problematic in our context, as $\hat{w}(t) = \frac{Q(t)}{\mu(t)}$ may significantly deviate from the true value of $w(t)$.

To tackle this problem, we employ an iterative approach inspired by Lu et al. (2023), as

illustrated in Fig. 5(b), to approximate the actual time-dependent waiting time $w(t)$. The iterative process is represented by Eq. (24), where the initial value is set as $\widehat{w}^0(t) = 0$. Here, $\widehat{w}^k(t)$ is the k-th approximation of $w(t)$, $R^k(t)$ represents the residue of $Q(t)$ at iteration k, and $D(t) = \int_{t_0}^{t} \mu(\tau)d\tau$ denotes the cumulative departure at time t. In each iteration, the departure rate that is used to update $\widehat{w}^k(t)$ corresponds to the bottleneck capacity Cap. The iterative formula is given as:

$$\widehat{w}^{k+1}(t) = \frac{R^k(t)}{Cap} + \widehat{w}^k = \frac{Q(t) - D(t + \widehat{w}^k(t)) + D(t)}{Cap} + \widehat{w}^k(t), k = 0,1,2,\dots \tag{24}$$

Eq. (24) outline the approximations for $w(t)$. Notably, $\widehat{w}^k(t)$ remains an analytical function of t throughout, enabling the choice of a suitable k to ensure the desired level of approximation accuracy.

For iteration 1:

$$\widehat{w}^1(t) = \frac{Q(t)}{Cap} \tag{25}$$

The final approximation, $\widehat{w}(t)$, is denoted as $\widehat{w}^2(t)$ for $k = 2$ in this study. That is, the time-dependent waiting time formula is given as:

$$w(t) = \frac{Q(t) - D(t + \frac{Q(t)}{Cap}) + D(t)}{Cap} + \frac{Q(t)}{Cap} \tag{26}$$

With this, the time-dependent travel time $tt(t)$ can be expressed as:

$$tt(t) = t_f + w(t) \tag{27}$$

It should be noted that $tt(t)$ represents the travel time for vehicles entering the link upstream at time t.

The inflow curvature parameter γ determines the shape of queueing function with respect to t, the shape of derived speed, and delay profiles. The total delay between the time t_0 and t_3 can also be calculated by the area between $A(t)$ and $D(t)$ in Fig. 5(a) through integrating $Q(t)$, so we have $W = \int_{t_0}^{t_3} Q(t)dt = \int_{t_0}^{t_3} \frac{\gamma}{3}(t - t_0)^2(t_3 - t)dt$. Again, this looks intermediating. Similarly, we will utilize the substitution method and let $u = \frac{t - t_0}{t_3 - t_0}$. We will have $dt = (t_3 - t_0)du$, $t - t_0 = u(t_3 - t_0)$, and $t_3 - t = (1 - u)(t_3 - t_0)$. Additionally, when $t = t_0$, $u = 0$; and when $t = t_3$, $u = 1$. In the end, we will obtain the following function. For detailed derivation, please refer to Appendix B.4.

$$W = \int_{t_0}^{t_3} Q(t)dt = \int_0^1 \frac{\gamma}{3}[u(t_3 - t_0)]^2 \cdot [(1-u)(t_3 - t_0)] \cdot (t_3 - t_0)du \tag{28}$$
$$= \int_{t_0}^{t_3} \frac{\gamma}{3}(t - t_0)^2(t_3 - t)dt = \frac{\gamma}{36}(t_3 - t_0)^4$$

If we denote the congestion duration as P, i.e., $P = t_3 - t_0$, then the total delay W could be rewritten as the following:

$$W = \frac{\gamma}{36}P^4 \tag{29}$$

4.3 Conditions for efficiency gains

In this subsection, we discuss how the developed FDQ-PN modeling framework can be utilized to mitigate efficiency losses and enhance overall efficiency. In addition, the FDQ-PW case is presented in Appendix C.

We use Fig. 6 to illustrate the efficiency loss and gain for the FDQ-PN model. Three subfigures are provided, each representing a different perspective: (a) demand side, (b) supply side, and (c) demand-supply coupled perspective. In each subfigure, as discussed in the methodology section above, both the arrival rate (red curve) and the throughput (blue curve) are modeled as cubic functions.

We're interested in quantifying how slight adjustments to arrival flow patterns (i.e., inflow control) can sustain improvements in throughput while reducing queue length and delays. This helps us better understand the mechanisms of efficiency loss and explore the potential for mitigating these losses through targeted inflow adjustments. By achieving a balance between inflow and outflow, even under heavy congestion, these adjustments can be effective. To achieve this, we conduct sensitivity analysis for various scenarios. In Fig. 6(a), we adjust the inflow rate while keeping the flow throughput fixed, which mimics real-world traffic management strategies like reservation and tolling. In Fig. 6(b), we maintain the inflow rate but adjust the throughput, which could result from ramp metering or one-way street conversions. Finally, in Fig. 6(c), both the inflow rate and throughput can be adjusted.

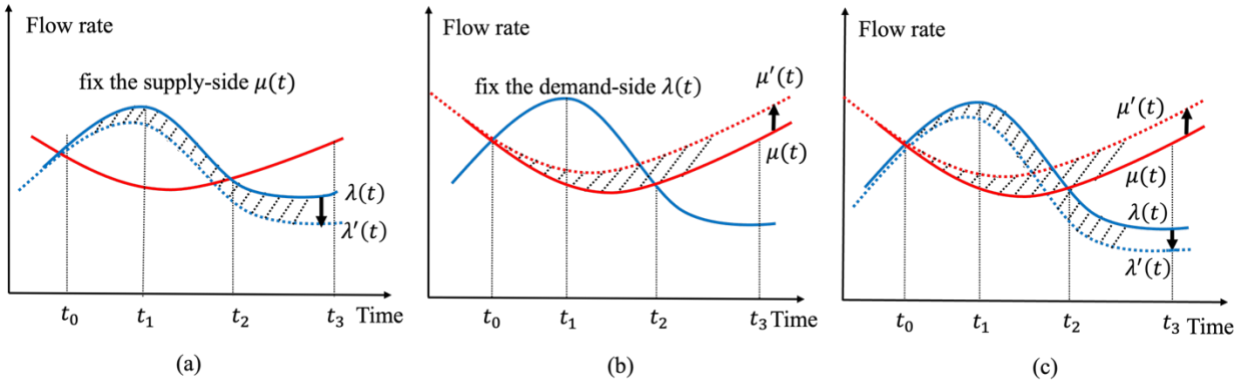


Figure 6 Illustration of different efficiency gain in the FDQ-PN: (a) demand-side

change strategy; (b) supply-side change strategy; (c) demand-supply coupled change strategy.

In the FDQ framework, the purpose of Section 4.3 is not to predict new onset or clearance times under hypothetical control strategies, but rather to evaluate how inflow smoothing or throughput enhancement, when applied within the same observed congestion window, affects queue evolution, delay, and throughput efficiency.

To ensure a consistent comparison baseline, all strategies are evaluated within the same time horizon $[t_0, t_3]$ derived from field data. Thus, modifications are applied directly to the net-flow curve rather than through re-estimating the turning points t_1 and t_2.

Conceptually, demand-supply adjustments may shift these timestamps under idealized conditions, but operationally, the FDQ simulations keep the timestamps fixed so that improvements in efficiency metrics can be compared under identical temporal boundaries. This distinction reconciles the narrative intuition and the algorithmic implementation.

Next, we define three primary indices for evaluating the efficiency loss and gain due to these adjustments, including the longest queue lengths Q_{max}, the total delay W and the average delay per vehicle $\overline{w}$, and will develop closed-from equations to quantify their values. These indices not only help to quantify the adverse impacts of traffic congestion but also provide insight into the potential benefits of implementing effective traffic management strategies to alleviate congestion.

(1) Longest queue length Q_{max}: indicates the peak level of congestion. Longer queues not only signify severe traffic bottlenecks but also lead to increased emissions, fuel consumption, and potential blockages of adjacent intersections or bottlenecks, exacerbating systemic inefficiency. Conversely, reducing Q_{max} can unlock potential efficiency gains by preventing spillover effects, improving throughput, and reducing environmental impacts, thereby enhancing the overall resilience of the traffic network. Since the longest queue is observed at time t_2, we have $Q_{max} = Q(t_2)$.

(2) Total delay W: reflects the cumulative time lost by all vehicles due to congestion. High W values correspond to significant efficiency losses, as vehicles and drivers experience prolonged idling and stop-and-go conditions. Implementing strategies to minimize W, such as demand-responsive tolling or real-time traffic rerouting, can yield efficiency gains by optimizing traffic flow, reducing delays, and enhancing the productivity of both travelers and the transportation network as a whole.

(3) The average delay per vehicle $\overline{w}$: represents the average delay experienced by each vehicle at the bottleneck. It can be calculated as the total delay divided by the total number of vehicles in the queue, expressed as: $\overline{w} = W / \sum_{t_0}^{t_3} Q(t)$.

We adhere to the work presented in (Newell 1982), in a small neighborhood around the peak congestion time t_1, the deviation between the arrival flow $\lambda(t)$ and the throughput $\lambda(t)$ can be locally approximated by a symmetric quadratic function centered at t_1:

$$\mu(t) = \lambda(t_1) - \gamma(t - t_1)^2 \tag{30}$$

If $\mu(t)$ is sufficiently close to $\lambda(t_1)$, at time t_0 we have $\mu(t) = \lambda(t_0)$ and at time t_2 where t_2 is again equal to $\lambda(t)$, $\mu(t) = \lambda(t_2)$, and using the fact that $\mu(t) = \mu(t_2) = \mu(t_1)$ yields.

$$\mu(t_1) = \lambda(t_1) - \gamma(t_0 - t_1)^2 \tag{31}$$

$$\mu(t_1) = \lambda(t_1) - \gamma(t_2 - t_1)^2 \tag{32}$$

Solving these two expressions directly and we could derive t_0 and t_2 with the following equation:

$$t_0 = t_1 - [\frac{\lambda(t_1) - \mu(t_1)}{\gamma}]^{1/2}, t_2 = t_1 + [\frac{\lambda(t_1) - \mu(t_1)}{\gamma}]^{1/2} \tag{33}$$

We define the Net Flow Volume Index (NFII) (Doll et al., 2024) to analyze the efficiency loss and gain in the FDQ model. The NFII is the normalized highest net flow volume, calculated as the difference between the highest arrival flow rate and the constant outflow rate. Mathematically, that is $NFII \sim [\frac{\lambda(t_1)-\mu(t_1)}{\gamma}]^{1/2}$. In the following, we derive the relationship between NFII and the key evaluation parameters, including congestion duration P, the longest queue lengths Q_{max} and total delay W.

The congestion duration, $P = t_3 - t_0$, should be proportional to the square root of the NFII. This relationship is analogous to the inverse of the curvature's second-order gradient γ.

$$P = 3[\frac{\lambda(t_1) - \mu(t_1)}{\gamma}]^{1/2} \tag{34}$$

The longest queue length, Q_{max}, should be proportional to the NFII, raised to the power of 3/2, expressed as $[\frac{\lambda(t_1)-\mu(t_1)}{\gamma}]^{3/2}$. This is akin to the inverse of the curvature's second-order gradient γ, $\gamma > 0$.

$$Q_{max} = \frac{\gamma}{3}[\frac{\lambda(t_1) - \mu(t_1)}{\gamma}]^{3/2} \tag{35}$$

The total delay per period W should be proportional to the NFII raised to the power of 2, expressed as $[\frac{\lambda(t_1)-\mu(t_1)}{\gamma}]^2$.

$$W = \frac{9\gamma}{4}[\frac{\lambda(t_1) - \mu(t_1)}{\gamma}]^2 \tag{36}$$

It is proportional to the square of the amount of oversaturation $\lambda(t_1) - \mu(t_1)$.

The detailed progress to analyzing efficiency loss and gain with the FDQ-PN model is shown in Algorithm 1.

Algorithm 1: Efficiency loss and gain analysis using FDQ-PN modeling approach

Step 1: Benchmark parameter calculation: Calibrate the initial benchmark parameters with the field data: net flow π, arrival rate at t_0, $\lambda(t_0)$, flow throughput at t_0, $\mu(t_0)$, arrival rate at t_2, $\lambda(t_2)$, flow throughput at t_2, $\mu(t_2)$, maximum queue length $Q(t_2)$, density at t_2, $k(t_2)$, backward wave speed w_b, total delay W, average delay $\bar{w}$.
Step 2: Reduced net flow calculation: The adjusted net flow π' is calculated by multiplying the original flow π by a reduction factor, which is expressed as a percentage.

(i) **Demand-side strategy:** The updated arrival rate is $\lambda'(t) = \lambda(t) - (\pi - \pi')$, while the flow throughput rate remains unchanged $\mu'(t) = \mu(t)$;
(ii) **Supply-side strategy:** The new flow throughput is given by $\mu'(t) = \mu(t) + (\pi - \pi')$, with the arrival rate remains unchanged $\lambda'(t) = \lambda(t)$;
(iii) **Demand-supply coupled strategy:** The queue length at t_1 is updated as $Q'(t_1) = Q(t_1) - [\lambda(t_1) - \lambda'(t_1)] \cdot \frac{1}{3}$, the flow throughput at t_1 is updated as $\mu'(t_1) = Cap - \theta \cdot Q'(t_1)$.

Step 3: Performance parameters in FDQ-PN calculation:

(i) The maximum queue length is estimated by ${Q_{max}}' = Q(t_2) = \frac{\gamma'}{3}[\frac{\lambda'(t_1) - \mu'(t_1)}{\gamma'}]^{3/2}$

(ii) The total delay is estimated by $W' = \frac{9\gamma'}{4}[\frac{\lambda'(t_1) - \mu'(t_1)}{\gamma'}]^2$

(iii) The average delay is estimated by $\overline{w}' = W' / \sum_{t_0}^{t_3} Q'(t)$

Step 4: Efficiency loss and gain analysis:

(i) Efficiency gain in total delay: Compare the total delay before and after implementing the strategy and the relative reduction in total delay is:

$$\Gamma_W = \frac{W - W'}{W} \times 100\%$$

(ii) Efficiency gain in congestion: Compare the average delay before and after implementing the strategy and the relative reduction in average delay is:

$$\Gamma_P = \frac{P - P'}{P} \times 100\%$$

(iii) Efficiency gain in average delay: Compare the average delay before and after implementing the strategy and the relative reduction in average delay is:

$$\Gamma_{\overline{w}} = \frac{\overline{w} - \overline{w}'}{\overline{w}} \times 100\%$$

5 Numerical Experiments

5.1 Traffic data overview and multi-step bottleneck identification

In this section, we use two datasets for model calibration and numerical analysis purposes. The first dataset, from PeMS Los Angeles (LA) I-405, includes traffic flow, speed, and occupancy data collected via loop detectors at 5-minute intervals throughout May 12, 2017. We apply the critical speed method to identify the congestion period in this section. The flow data are already provided in passenger-car-equivalent units, and heavy-vehicle percentages at the site are low enough that their impact on FD estimation and throughput calibration is negligible.

The congestion period occurred from 12:20 (i.e., $t_0 = 12.33$) to 20:20 (i.e., $t_3 = 20.33$) at a bottleneck located at Absolute Postmile (Abs) 13.74 (Detector 4 in Fig. 11), as shown in Fig. 7(a). The second dataset, from Phoenix I-10, features traffic flow and speed data collected via remote traffic microwave sensors at 5-minute intervals throughout January 5, 2016. The congestion period occurred from 14:10 (i.e., $t_0 = 14.17$) to 19:10 (i.e., $t_3 = 19.17$) at a bottleneck near downtown L139, as shown in Fig. 7(b). Both bottlenecks are common issues caused by morning and evening peak periods. To ensure that the calibration is not overly dependent on a single day of observations, we additionally provide cross-day robustness checks (Section 5.3.4), which serve as an out-of-sample validation of the FDQ model.

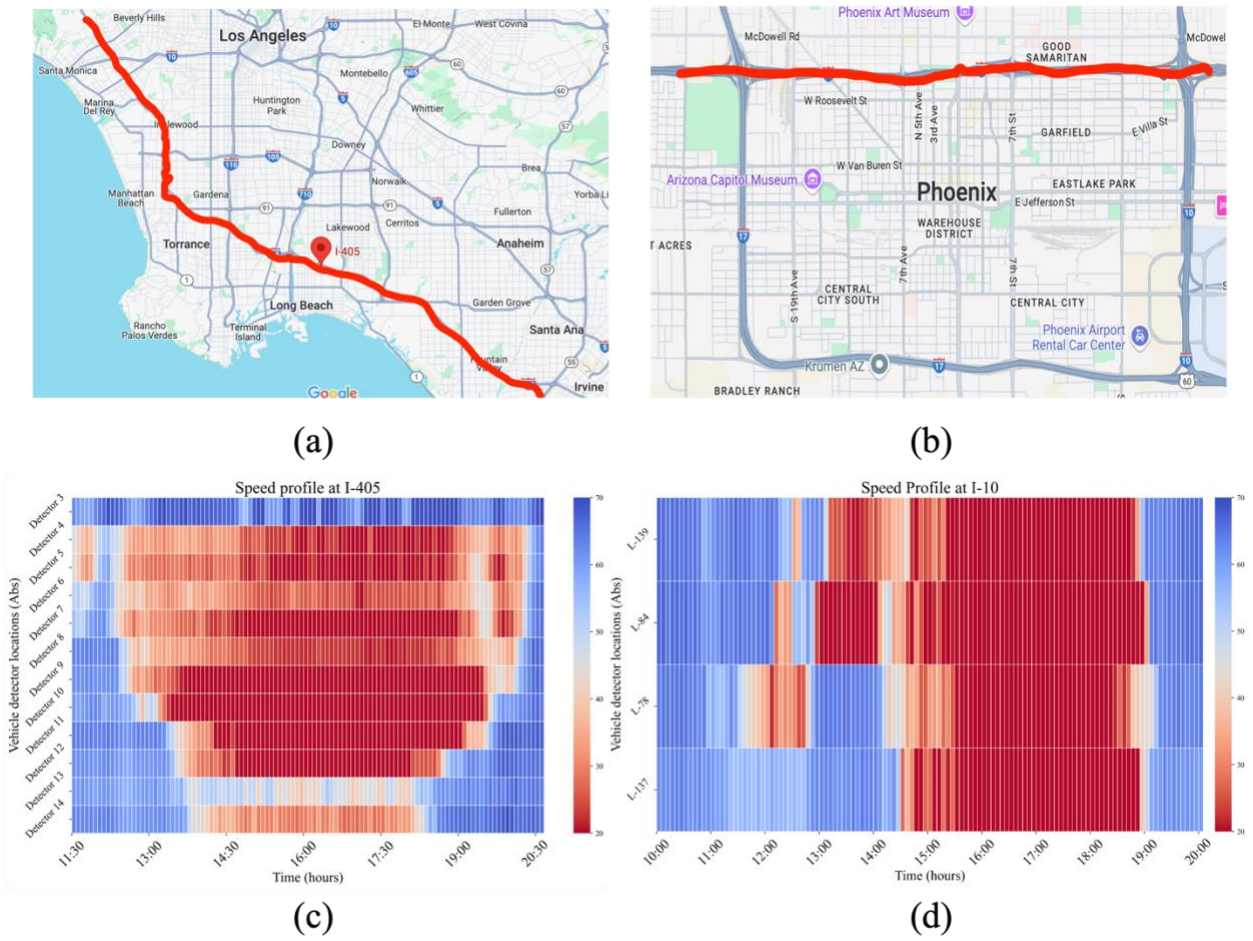


Figure 7 Areas of selected corridors used in the two datasets: (a) Los Angeles I-405 corridor; (b) Phoenix I-10 corridor; (c) speed heatmap of I-405 corridor; (d) speed heatmap of I-10 corridor.

To start off, we clarify that in the LA I-405 corridor PeMS dataset, the detector is located within the bottleneck area and provides an accurate representation and approximation of the queue propagation and dissipation process. The relationship between Detectors 3 and 4 is shown in Fig. 8(a). The traffic bottleneck selected for this case study is monitored by Detector 4, which is situated at milepost 13.74, within an active bottleneck area near Brookhurst Street. Detector 3 is situated downstream of this location. The associated on-ramp is located at milepost 13.82, and the off-ramp at milepost 13.37, with supporting evidence provided in Fig. 8(b) and Fig. 8(c). These numbers and screenshots were obtained from the Caltrans Postmile Query Tool provided by Caltrans Postmile Services. This detector is approximately 0.08 miles downstream from the ramp merge point and is considered one of the closest documented detectors in the literature capable of capturing traffic conditions within the bottleneck region.

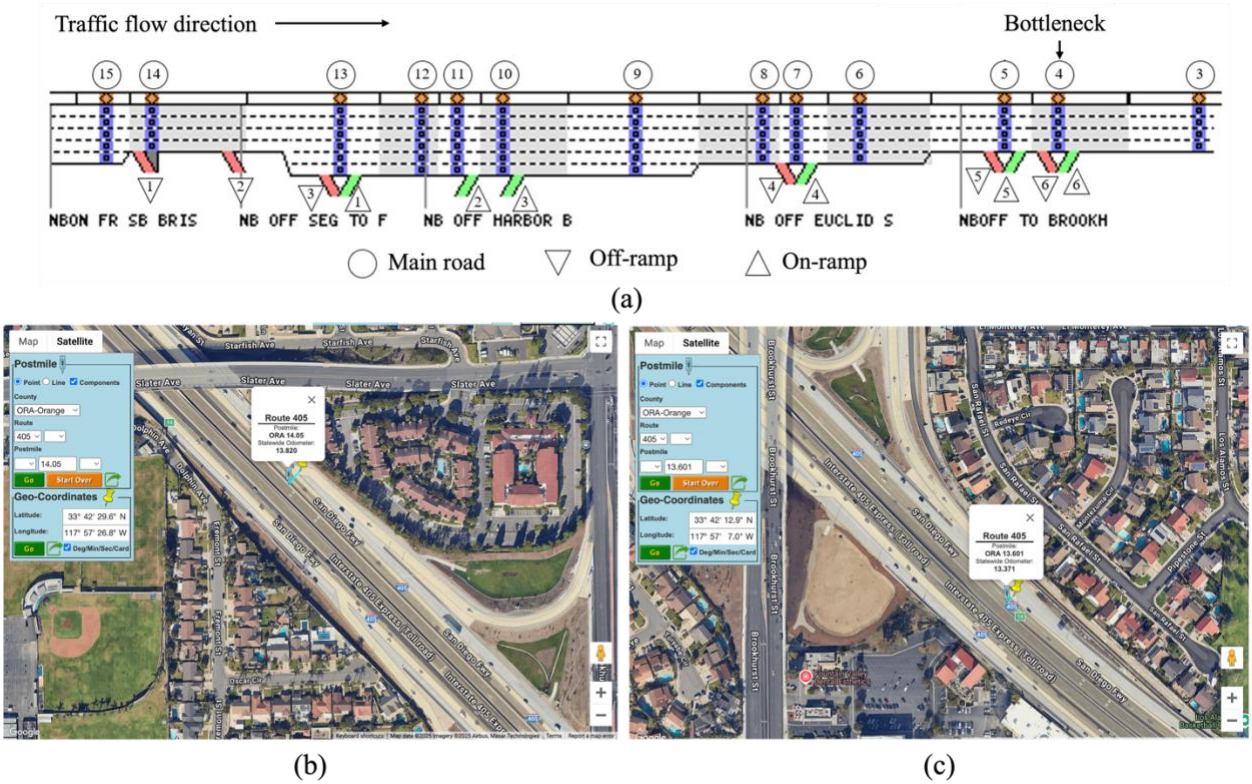


Figure 8 Illustration of detector location in Los Angeles I-405 corridor: (a) bottleneck area (source: PeMS) (b) on-ramp of the bottleneck, and (c) off-ramp of the bottleneck (source: Caltrans online postmile query tool).

Fig. 9 provides a comprehensive analysis of traffic conditions at Detector 4, confirming the presence of a persistent bottleneck with recurring congestion. The top row illustrates key fundamental relationships: (a) the flow-density plot reveals a clear distinction between uncongested (blue) and over-congested (red) states; (b) the flow-speed plot further differentiates these regimes, showing sharp speed reductions under similar flow levels in the congested phase; and (c) the speed-density curve captures a rapid speed decline once density exceeds the critical threshold. The bottom row presents time-series patterns: (d) flow exhibits a plateau followed by a decline during peak hours; (e) speed drops significantly from morning to afternoon and remains low, indicating sustained congestion; and (f) density rises steeply during peak periods, exceeding the critical value. Collectively, these observations confirm a well-defined over-congested state at Detector 4, marked by reduced flow, low speeds, and high densities.

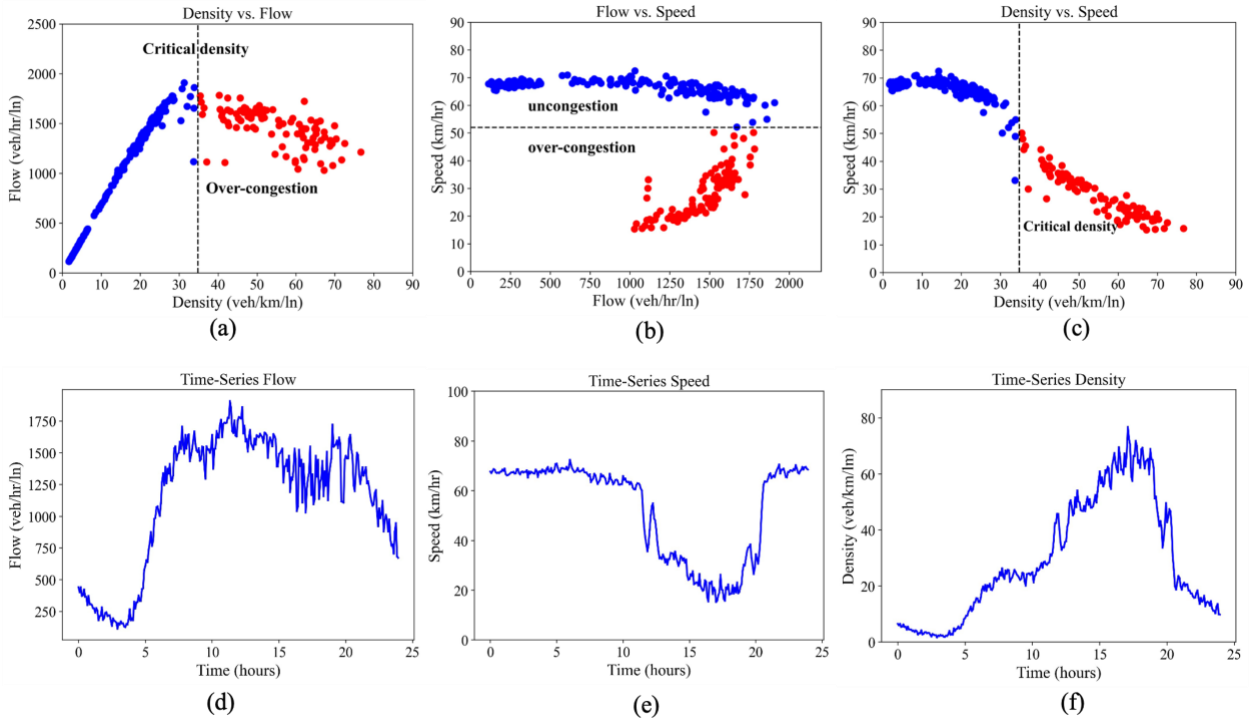


Figure 9 FD and time series diagrams at the downstream bottleneck in Detector 4.

To confirm the absence of downstream congestion, Fig. 10 presents traffic data from Detector 3, showing no signs of an active bottleneck or flow restriction. The top row displays fundamental relationships under uncongested conditions: (a) flow increases with density without reaching saturation, (b) speed remains stable across varying flow levels, indicating smooth operations, and (c) high speeds persist at low densities, confirming free-flow conditions. The bottom row illustrates 24-hour time-series trends: (d) flow follows typical morning and evening peaks, (e) speed shows only minor declines during peak periods, and (f) density fluctuates with demand but remains within uncongested ranges. Together, these patterns confirm that no significant congestion or flow reduction occurred at Detector 3, and a persistent bottleneck had not formed at this location.

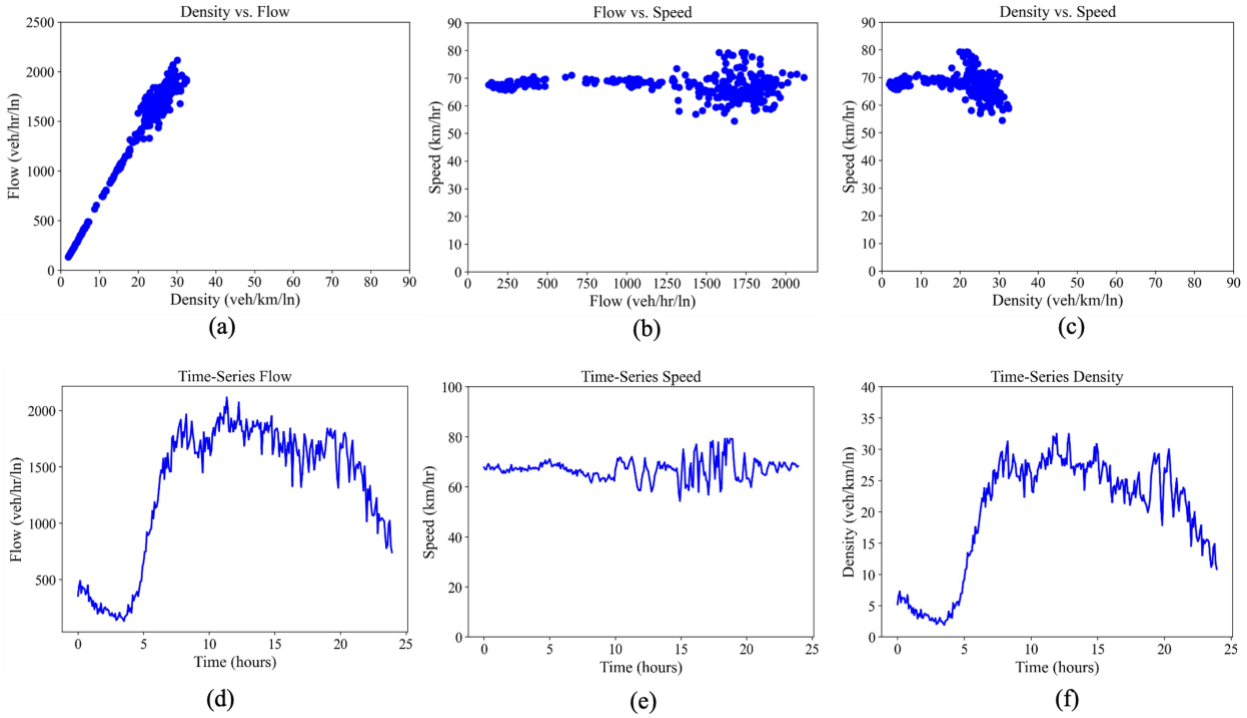


Figure 10 FD and time series diagrams at the downstream bottleneck in Detector 3.

To complement the speed-based bottleneck identification, we incorporate a transform-based analysis using cumulative arrival and departure curves, following established methods in traffic flow literature (Cassidy and Windover, 1995; Windover and Cassidy, 2001). As demonstrated by Windover and Cassidy (2001), Bertini and Cassidy (2002), and Chung and Cassidy (2002), these transformed curves effectively reveal active bottlenecks by highlighting upstream accumulations and downstream free-flow conditions. This analysis is applied to 5-minute loop detector data from Detectors 3, 4, and 5, as shown in Fig. 8.

The results obtained by the above-mentioned transform-based method are shown in Fig. 11. It illustrates the evolution of traffic flow at the targeted bottleneck segment. Three detectors (i.e., detectors 3, 4, and 5 in Fig. 8(a)) are selected upstream and downstream of the identified bottleneck. Fig. 11(a) presents the original time-series flow data, revealing significant fluctuations during the congested period, with a characteristic "decrease-then-increase" pattern. Fig. 11(b) shows the cumulative arrival curves, which visually represent the cumulative counts collected directly from the freeway. To better capture the temporal shift between upstream and downstream observations, the background flow is subtracted. As shown in Fig. 11(c), applying a background flow reduction amplifies flow state changes on the cumulative curves, making disturbances more visually identifiable. Fig. 11(d) presents the queue length estimated as the vertical difference between the moving-time-adjusted cumulative arrival curves from Detectors 3 and 4. The resulting time series shows a clear "increase-then-decrease" trend in queue length.

From the Fig. 11, it is observed that stable congestion begins around 12:20 and dissipates by 18:20. Outside this stable period, traffic exhibits transitional oscillations, with generally steady demand punctuated by local fluctuations.

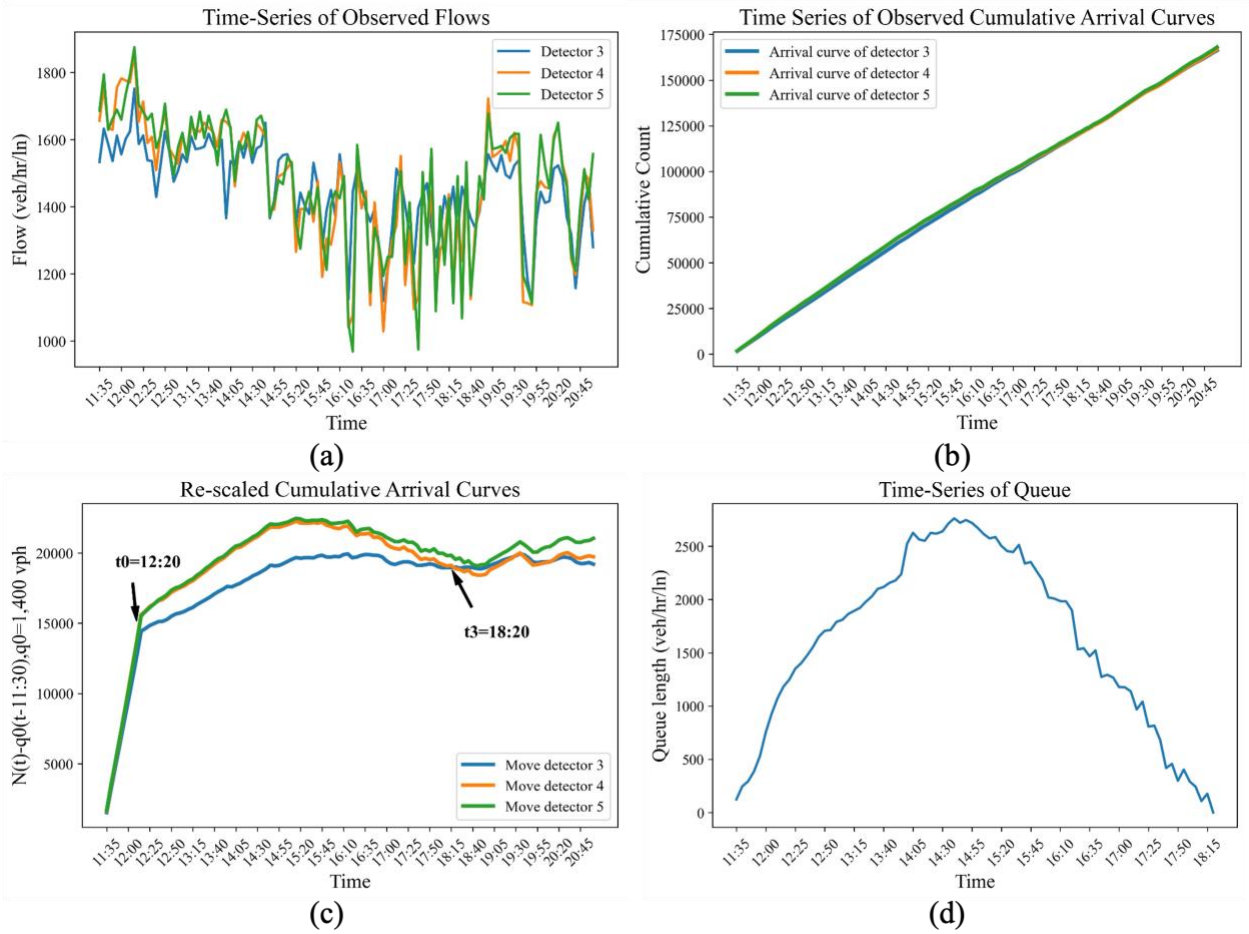


Figure 11 Empirical flow and queue dynamics for transformer-based method.

5.2 Calibration of fundamental diagram parameters

Triangular fundamental diagram was plotted to estimate bottleneck capacity based on flow versus density scatterplots (Dervisoglu et al., 2009). First, as shown in Fig. 12(a), the maximum capacity was estimated by averaging the five highest flow values across a segment within a simulated period of 5 minutes. Next, the free-flow speed was determined as the slope of a regression line fitted to points with speed values exceeding the speed limit. The intersection's vertically projected value was defined as the critical density, beyond which traffic entered a congested state.

When the bottleneck was over-congested, the actual bottleneck capacity could vary depending on the bottleneck's severity. During congestion, the bottleneck capacity (Cap) was identified as the flow value at the critical density along the least-squares regression line of points with density values above the critical density. It refers to the post-drop bottleneck capacity unless otherwise stated. The slope of the line in the congested regime represents the backward wave speed, while the minimum flow rate during congestion indicates the lowest sustained flow under these conditions.

The calibration method employed in this study involves collecting traffic flow data from observation equipment and using the least squares method (LSM) to calibrate the triangular FD

model. The primary aim is to minimize the disparity between the observed flow q_i data and the estimated flow data $\hat{q}$ using the triangular FD model.

$$min\ Z = \sum_{i=1}^{n} [q_i - \hat{q}(k_i; v_f, k_c, k_j, w_b)]^2 \tag{37}$$

where v_f, k_c, k_j, w_b are the parameters for calibration. n is the total amount of data, q_i and k_i are the observed flow and density data.

The calibration results in the selected two datasets are shown in Fig. 12. The green dots in the figure represent the observed flow and density data, while the solid red lines represent the fitted curves using the triangular FD. The calibrated parameters are listed in Table 1. The calibrated errors are listed in Table 2.

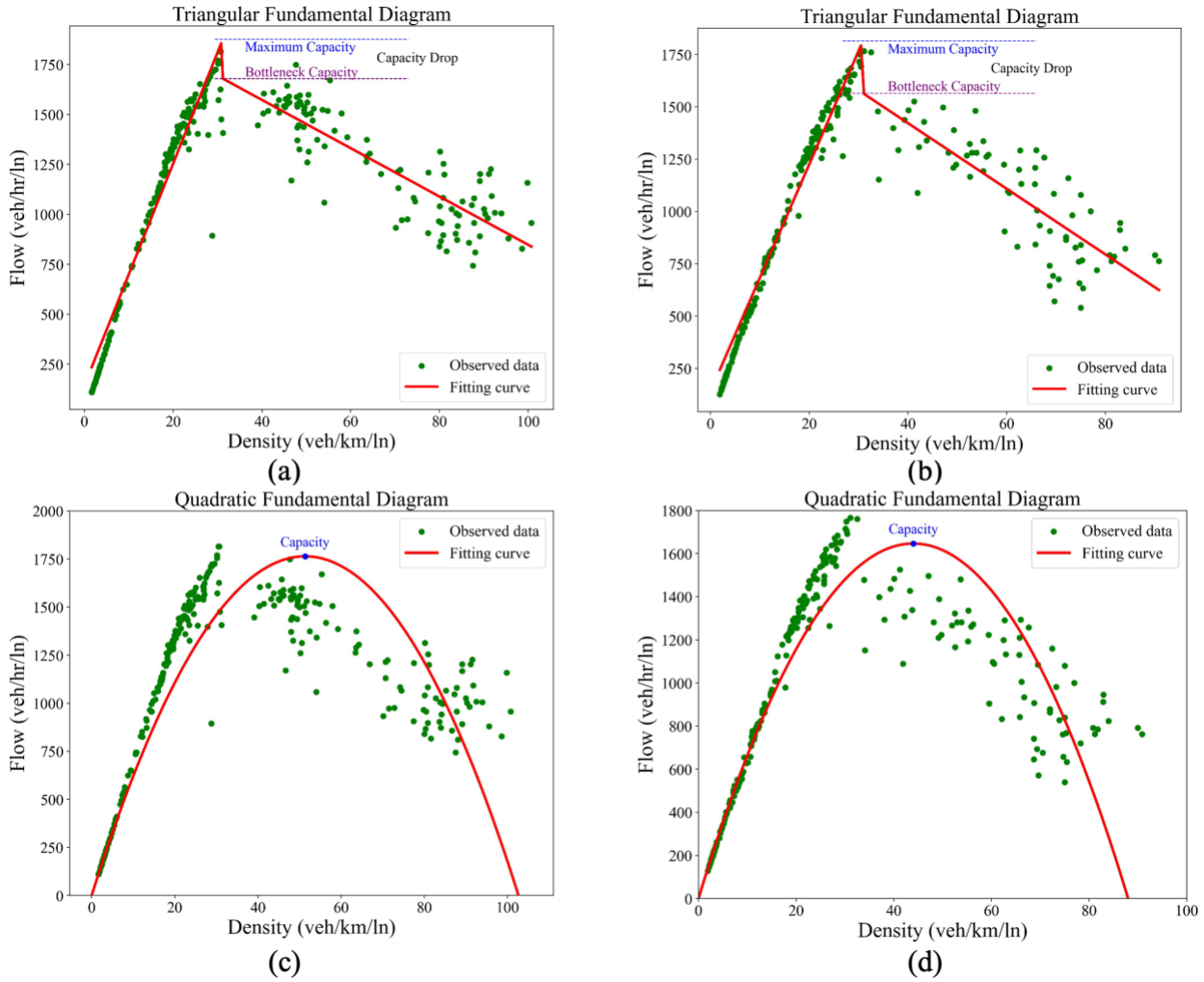


Figure 12 Analysis of FD calibration with observation data (a) triangular FD calibration of LA I-405 dataset; (b) triangular FD calibration of Phoenix I-10 dataset; (c) quadratic FD calibration of LA I-405 dataset: (d) quadratic FD calibration of Phoenix I-10 dataset.

Table 1 Calibrated parameters result with empirical dataset.

Parameters	Dataset 1: LA I-405 corridor	Dataset 2: Phoenix I-10 corridor
Bottleneck capacity Cap (veh/hr/ln)	1,679	1,564
Free flow speed v_f (km/hr)	67	68
Backward wave speed w_b (km/hr)	12.09	15.68
Critical speed v_c (km/hr)	53.92	50.57
Critical density k_c (veh/km/ln)	31.14	30.93
Jam density k_j (veh/km/ln)	158.10	121.69
Inflow curvature γ	1.71	5.37
Queue impact coefficient θ	14.51	34.54

In LA dataset, bottleneck capacity is 1,679 veh/hr/ln, the free-flow speed is 67 km/hr, representing typical conditions in the absence of congestion. The backward wave speed, which characterizes the propagation of congestion, is 12.09 km/hr. The critical density at the onset of congestion is 31.14 veh/km/ln, while the jam density, representing the density in fully congested traffic, reaches 158.10 veh/km/ln. These values indicate that the LA corridor exhibits relatively high density thresholds and can sustain larger volumes before reaching breakdown.

In the Phoenix dataset, the bottleneck capacity is slightly lower at 1,564 veh/hr/ln. The free-flow speed is 68 km/hr, marginally higher than in the LA dataset, but the backward wave speed is substantially higher at 15.68 km/hr, indicating faster propagation of congestion. The critical density is comparable to that of the LA dataset at 30.93 veh/km/ln, whereas the jam density is notably lower at 121.69 veh/km/ln, suggesting a reduced maximum density under fully congested conditions. The higher jam density observed in the LA dataset implies that it can accommodate more vehicles during severe congestion, although this may also lead to longer or more persistent queues.

Overall, Dataset 1 (LA) exhibits a higher capacity, higher jam density, and lower congestion-propagation speed, suggesting greater ability to carry traffic under both free-flow and congested regimes. In contrast, Dataset 2 (Phoenix) shows faster backward wave propagation and a lower jam density, implying quicker congestion spread and lower storage capability once congestion forms. These differences highlight that the LA corridor tends to sustain higher volumes and accumulate larger queues, whereas the Phoenix corridor experiences more rapidly evolving but lower-density congestion.

Table 2 Compare the parameter calibration error between observed data and estimated data.

Dataset	Models	MAE	RMSE	MAPE	R^2

Dataset 1: LA I-405 corridor	Triangular	68.13	106.13	6.07	0.95
	Quadratic	192.73	243.28	15.84	0.75
Dataset 2: Phoenix I-10 corridor	Triangular	51.28	86.34	5.17	0.95
	Quadratic	125.07	191.49	12.40	0.84

Table 2 reports the calibration errors between the observed and estimated values for both datasets, using mean absolute error (MAE), root mean square error (RMSE), mean absolute percentage error (MAPE), and coefficient of determination (R^2) as evaluation metrics. Across all measures, the Phoenix I-10 dataset presents smaller errors than the LA I-405 dataset. For the triangular FD model, Dataset 2 records MAE and RMSE values of 51.28 and 86.34, both lower than the 68.13 and 106.13 in Dataset 1. Its MAPE also decreases from 6.07 percent in Dataset 1 to 5.17 percent in Dataset 2. Both datasets achieve an R^2 value of 0.95, indicating that the triangular model reproduces the observed traffic states with strong consistency.

In contrast, the quadratic FD performs significantly worse on both corridors. In Dataset 1, the MAE and RMSE increase to 192.73 and 243.28, nearly three times the errors of the triangular model, and the MAPE rises to 15.84 percent. The R^2 value drops to 0.75. In Dataset 2, although the quadratic model produces smaller errors than in Dataset 1, its MAE of 125.07, RMSE of 191.49, and MAPE of 12.40 percent remain substantially higher than the triangular model. Its R^2 value, 0.84, is also noticeably lower.

Overall, the quadratic form consistently yields larger errors and weaker goodness of fit across both corridors. Given its inferior performance, the subsequent section analyses focus on the triangular representation when evaluating additional performance aspects of the FDQ model.

5.3 Validation of FDQ model accuracy in traffic state estimation

In this section, we compare the errors in traffic state estimation between the benchmark models and the proposed FDQ models. For comparison purpose, two benchmark models are adopted. The first benchmark model utilizes a piecewise arrival rate and a constant departure rate and is thus referred to as the "PW" model. The second benchmark model is based on the PAQ, which incorporates a polynomial arrival rate and constant departure rate. We then examine the modeling accuracy of the FDQ-PN and FDQ-PW model as well. The key equations for these models are shown in Table 3.

Table 3 Comparison of the difference of those models.

Models	Arrival rates	Departure rates	Unknown parameters	FD Consistent?
Benchmark Model 1: PW	Piecewise: $\begin{cases} \pi_1 + \frac{\sum_{t_0}^{t_3} q(t)}{t_3 - t_0}, t\epsilon(t_0, t_2) \\ \pi_2 + \frac{\sum_{t_0}^{t_3} q(t)}{t_3 - t_0}, t\epsilon(t_2, t_3) \end{cases}$	Constant: $\frac{\sum_{t_0}^{t_3} q(t)}{t_3 - t_0}, t\epsilon(t_0, t_3)$	π_1, π_2	No

Benchmark Model 2: PAQ	Polynomial: $\gamma(t-t_0)(t_2-t) + \frac{\sum_{t_0}^{t_3} q(t)}{t_3-t_0}, t\epsilon(t_0,t_3)$	Constant: $\frac{\sum_{t_0}^{t_3} q(t)}{t_3-t_0}, t\epsilon(t_0,t_3)$	γ	No
Model 3: FDQ-PW	Piecewise: Eq. (38)	Piecewise: Eq. (37)	$\pi_1, \pi_2, Cap, \theta$	Yes
Model 4: FDQ-PN	Polynomial: Eq. (21)	Polynomial: Eq. (20)	γ, Cap, θ	Yes

For the calibration method in this section, for benchmark model 1 (PW), we use the LSM method to calculate the estimated values and compare them with the observed data. For FDQ-PW, the unknown parameters are the bottleneck capacity, Cap, and the backward wave speed, w_b, which can be derived from the calibrated triangular FD in Section 5.2. The LSM method is then used to calculate the error between the estimated and observed data. In PAQ and FDQ-PN, the key inflow curvature parameter is γ, which was estimated using the following objective function.

$$minY = \sum_{t=1}^{|T|} [Y(t) - \hat{Y}(t)]^2 \tag{38}$$

where $|T|$ is the number of time intervals in the congestion period, $Y(t)$ and $\hat{Y}(t)$ are the estimated values and observations at time interval t, respectively. $Y(t)$ and $\hat{Y}(t)$ can be either outflow rate and/or queue length depending on the data availability.

5.3.1 Accuracy in arrival and departure rates estimation

We tested the above four methods on two datasets and plotted the arrival and departure rates, as shown in Fig. 13. The different arrival and departure rate curves of each method are clearly shown. The solid red line represents the flow throughput curve, and the solid blue line represents the arrival rate curve.

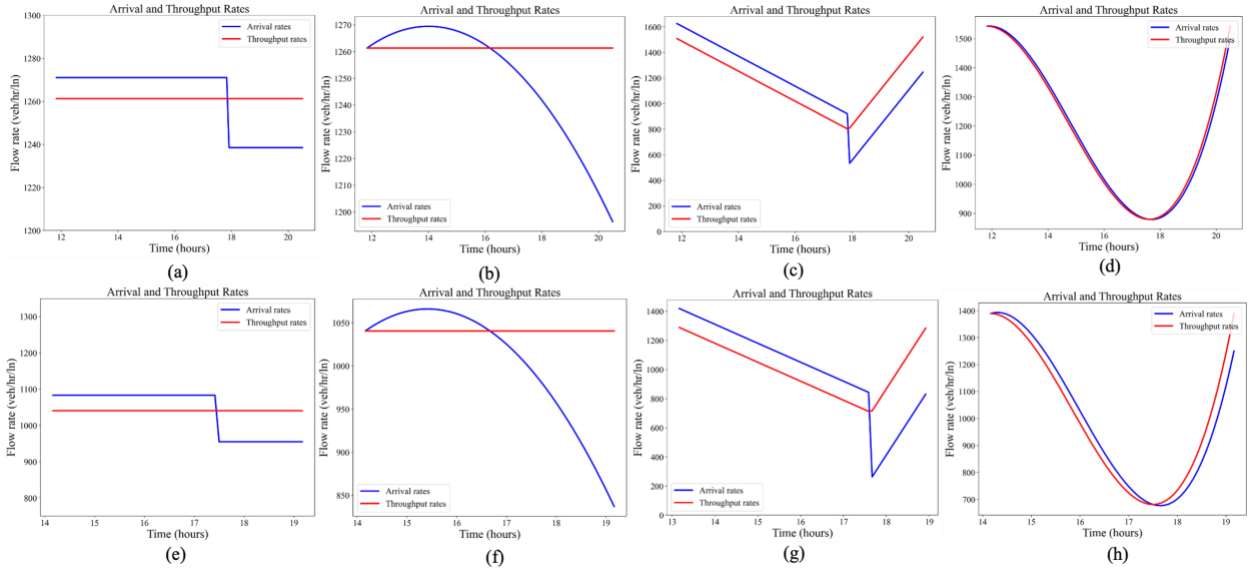


Figure 13 Arrival and flow throughput curves for the LA and Phoenix datasets: (a) and (e) Benchmark model 1: PW; (b) and (f) Benchmark model 2: PAQ; (c) and (g) Proposed model 1: FDQ-PW; (d) and (h) Proposed model 2: FDQ-PN; (The four graphs on the upper: LA case, four graphs on the lower: Phoenix case).

We used the observational data to fit the models listed in Table 3. As shown in Figs. 14 and 15, for the Los Angeles and Phoenix cases, respectively, the comparison methods include a constant throughput, a piecewise linear throughput, and a polynomial function throughput. The figures demonstrate that the polynomial departure rate method provides the best fit to the data, further validating our hypothesis. In this study, we used RMSE and MAPE as evaluation metrics. The specific error results for model fitting are presented in Table 4, where it is evident that the two FDQ models achieve much better performance when compared with the two benchmark models, and the FDQ-PN model is shown to have the lowest error among the four models. The two benchmark models have the same performance since they both assume a constant outflow rate.

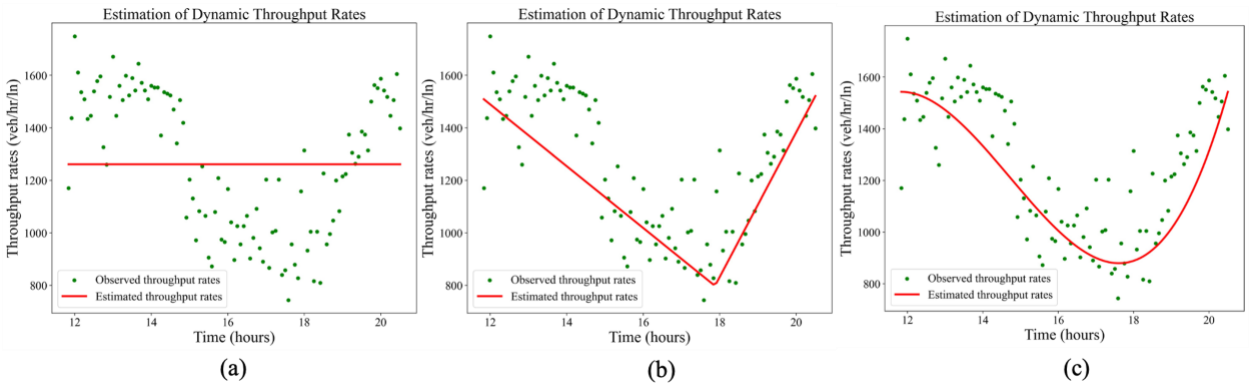


Figure 14 Fitting of flow throughput with different curves in the LA dataset: (a) Constant outflow rates; (b) Piecewise linear outflow rates; (c) Polynomial outflow rates.

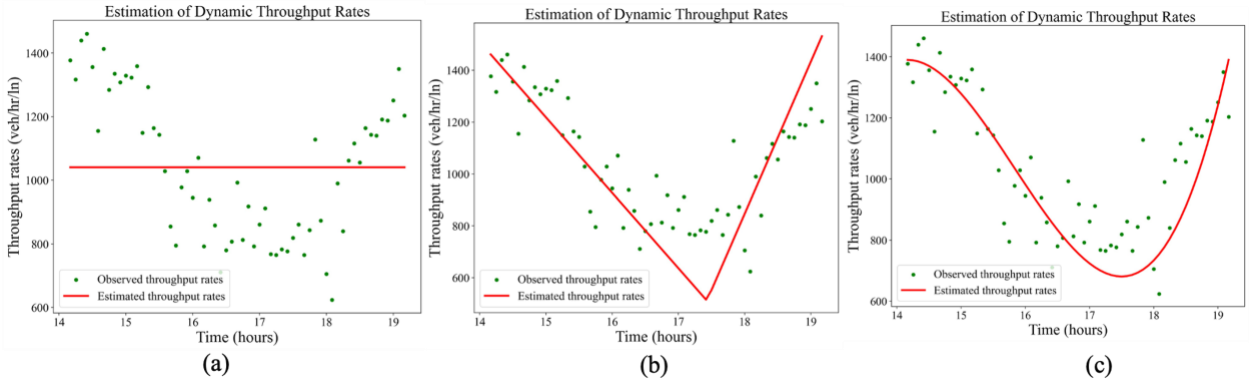


Figure 15 Fitting of flow throughput with different curves in the Phoenix dataset: (a) Constant outflow rates; (b) Piecewise linear outflow rates; (c) Polynomial outflow rates.

Table 4 Compare the flow throughput estimation error between the proposed FDQ model and the benchmark model.

Case	Dataset 1: LA I-405 corridor				Dataset 2: Phoenix I-10 corridor			
Models	PW	PAQ	FDQ-PW	FDQ-PN	PW	PAQ	FDQ-PW	FDQ-PN
RMSE	263.37	263.37	178.60	**169.57**	198.19	198.19	157.16	**131.26**
MAPE	20.17%	20.17%	10.89%	**10.86%**	20.16%	20.16%	13.49%	**10.57%**

We also calculated the Improvement (%) of the FDQ-PN model relative to the other models, as shown in the following. The FDQ-PN model outperformed the other three models across all evaluation metrics. The results are shown in Table 5.

$$Improvement(i) = \frac{baseline(i) - DTQ}{baseline(i)} \cdot 100\% \tag{39}$$

where $baseline(i)$ is the error of baseline model i, FDQ is the error of the FDQ-PN model.

Table 5 Improvement in accuracy of the FDQ-PN model over the benchmark model.

Case	Dataset 1: LA I-405 corridor		Dataset 2: Phoenix I-10 corridor	
Improvement (%)	Constant throughput	Piecewise linear throughput	Constant throughput	Piecewise linear throughput
RMSE	35.62	**5.06**	33.77	**16.48**
MAPE	46.16	**0.30**	47.57	**21.65**

To summarize, the FDQ models demonstrated high prediction accuracy and goodness of fit in both datasets, indicating its strong applicability in traffic flow prediction. As shown in Table 5, for dataset 1, the improvements compared with constant throughputs are as follows: RMSE improves by 35.62%, and MAPE improves by 46.16%. Compared with piecewise linear throughput, the improvements are 5.06% for RMSE, and 0.30% for MAPE. For dataset 2, the improvements under constant throughputs are: RMSE improves by 33.77%, and MAPE improves by 47.57%. Compared with piecewise linear throughputs, the improvements are 16.48% for RMSE, and 21.65% for MAPE.

5.3.2 Accuracy in queue length estimation

In this section, we analyze the accuracy of FDQ model in estimating the queue length profile. We fit the queue length as shown in Fig. 16. The estimation results are shown in Table 6. Across both datasets, the two proposed FDQ models generally performed better on these datasets compared to the two benchmark models for queue estimation, with FDQ-PW model having the lowest RMSE values. The two benchmark models have the same performance since they both assume a constant throughput. The results show that the proposed method can fit the queue length well and can represent the dynamic change of the queue.

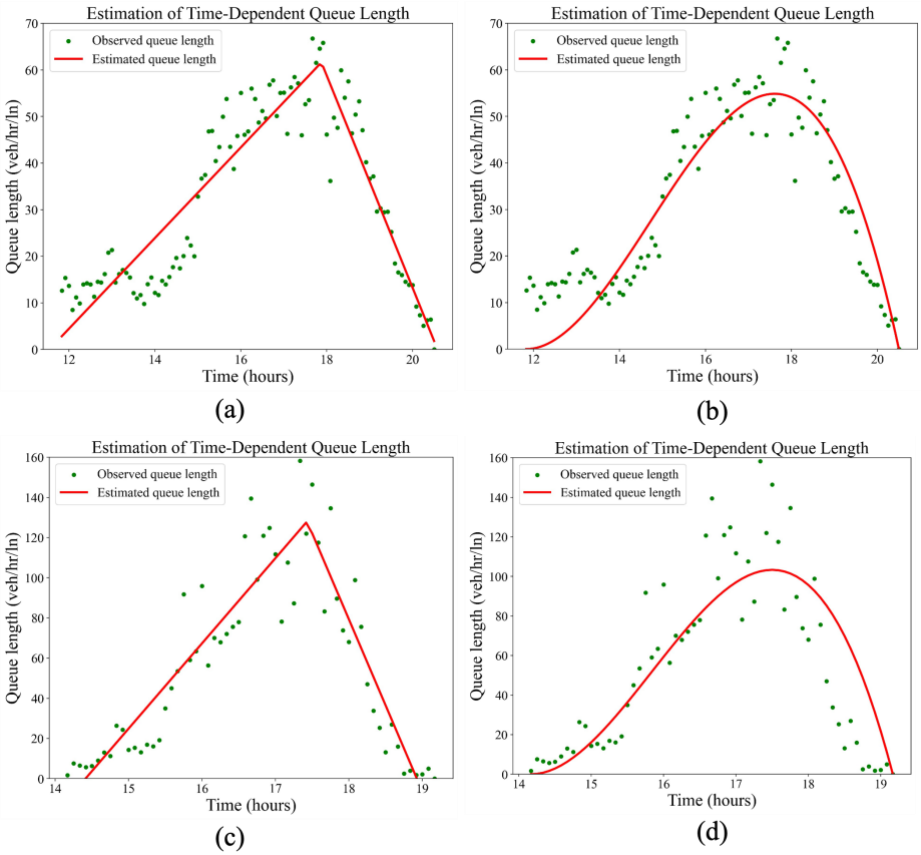


Figure 16 Fitting of queue length: (a) FDQ-PW in LA dataset; (b) FDQ-PW in LA dataset; (c) FDQ-PW in Phoenix dataset; (d) FDQ-PW in Phoenix dataset.

Table 6 Compare the queue estimation error between the proposed model and the benchmark model.

Case	Dataset 1: LA I-405 corridor				Dataset 2: Phoenix I-10 corridor			
Models	PW	PAQ	FDQ-PW	FDQ-PN	PW	PAQ	FDQ-PW	FDQ-PN
RMSE	7.83	7.83	**6.96**	7.81	24.79	24.79	**17.33**	24.72

5.3.3 Accuracy in speed and travel time estimation

In this section, we analyze the accuracy of FDQ-PN model in estimating the speed and travel time profile. We fit the speed as shown in Fig. 17. During congestion period, the estimated speed is obtained by travel time, during uncongested period, the estimated speed equals to the calibration free-flow speed v_f in Table 1. Table 7 compares the fitting speed error between observed and estimated data across two datasets. LA dataset shows lower MAE (4.62), RMSE (8.81), and MAPE (13.59%) compared to Phoenix dataset, which exhibits higher errors (MAE: 5.48, RMSE: 9.82, MAPE: 16.84%). However, dataset 1 achieves a higher R^2 value (0.84), indicating better model fit, compared to dataset 2's R^2 of 0.78. The proposed FDQ model is demonstrated to accurately estimate traffic speed in both cases.

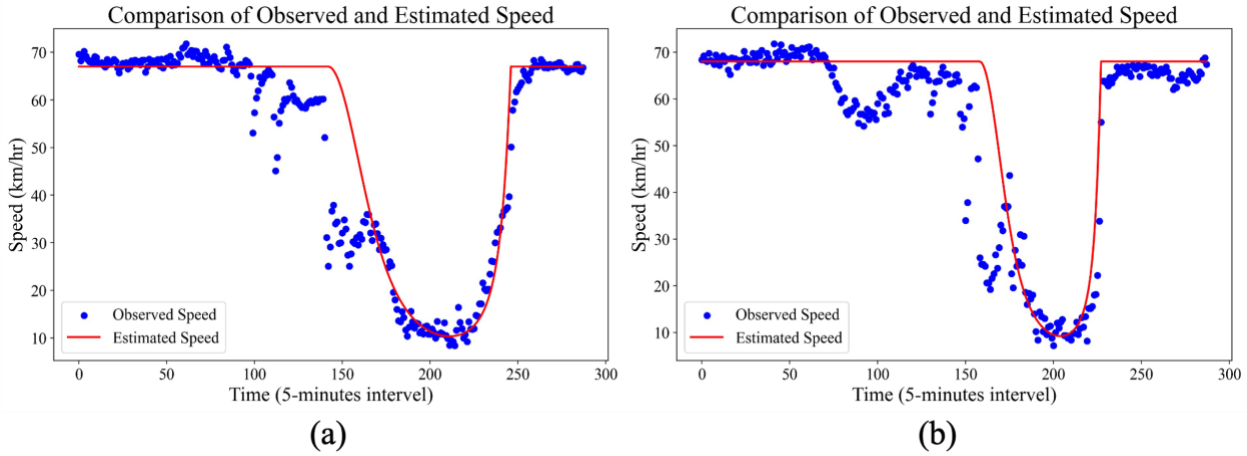


Figure 17 Estimated and observed speed curves: (a) LA dataset; (b) Phoenix dataset.

Table 7 Compare the fitting speed error between observed data and estimated data.

Case	MAE	RMSE	MAPE	R^2
Dataset 1: LA I-405 corridor	4.62	8.81	13.59%	0.84
Dataset 2: Phoenix I-10 corridor	5.48	9.82	16.84%	0.78

We fit the travel time as shown in Fig. 18, in which the estimated travel time are obtained by Eq. (27) in congestion period. When we estimate the travel time in uncongested period, the waiting time in travel time function is equal to 0. Then the estimated travel time equal the free-flow travel time t_f, $t_f = 1/v_f$. The estimation results are shown in Table 8. For Los Angeles dataset, the MAE is 14.17, RMSE is 26.08, MAPE is 9.96%, and R^2 is 0.92. For Phoenix dataset, the MAE is 16.32, RMSE is 34.79, MAPE is 10.98%, and R^2 is 0.87. The results indicate that the model achieves better fitting performance for the Los Angeles dataset than for the Phoenix dataset. The error measures MAE, RMSE, and MAPE are all lower for Los Angeles, and the corresponding R^2 value is higher, which demonstrates that the model captures the traffic dynamics more accurately in the Los Angeles case. In summary, the proposed FDQ model is demonstrated to accurately estimate travel time profile in both cases.

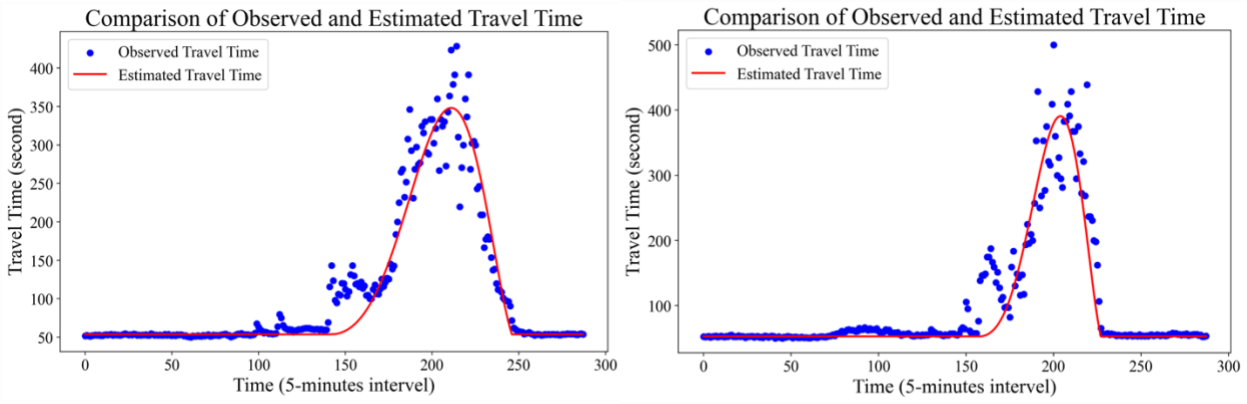


Figure 18 Estimated travel time curve and observed travel time with different cases: (a) LA dataset; (b) Phoenix dataset.

Table 8 Compare the fitting travel time error between observed data and estimated data.

Case	MAE	RMSE	MAPE	R^2
Dataset 1: LA I-405 corridor	14.17	26.08	9.96%	0.92
Dataset 2: Phoenix I-10 corridor	16.32	34.79	10.98%	0.87

5.3.4 Cross-day robustness analysis

To evaluate the day-to-day robustness of the calibrated FDQ model, we used ten weekday datasets collected from the same bottleneck along the I-405 corridor. All samples were selected within the same month to ensure comparable demand patterns and recurrent congestion conditions. The dataset was divided into two parts: The 80% data were used for model calibration, where the FDQ procedures described in Sections 5.1-5.3 were independently applied to estimate the FD parameters, the dynamic throughput function $\mu(t)$, and the queue evolution $Q(t)$.

The remaining 20% data were used for validation, where the averaged parameters obtained from the calibration period were applied to predict $\mu(t)$ and $Q(t)$ and compared with observations.

Table 9 summarizes the cross-day calibration and validation results. First, the fundamental diagram exhibits highly consistent behavior across days. Although the congestion period, $[t_0, t_3]$, fluctuate by about 1-3 hours due to normal peak-period variations, the MAPE remains consistently low at approximately 3.6%-5.7%. This persistently low relative error indicates that the flow-density relationship at this bottleneck is strongly recurrent. RMSE ranges between 62 and 92 veh/hr/ln, but the stable and low MAPE values better capture the robustness of the calibrated parameters.

The estimation of the throughput function $\mu(t)$ also demonstrates strong stability. While RMSE varies from 107 to 155 veh/hr/ln across days, the MAPE remains tightly bounded between 5.3% and 9.9%, suggesting that relative errors are well-controlled despite natural fluctuations caused by lane-changing intensity, merging turbulence, and temporary flow interruptions. When

the calibrated parameters were applied to the two validation days, $\mu(t)$ predictions maintained similar MAPE levels, confirming good generalizability of the FDQ model.

In contrast, queue length $Q(t)$ is more sensitive to absolute deviations, making RMSE the primary indicator of performance. Across the calibration period, the RMSE of $Q(t)$ remains low, between 4.86 and 8.31, showing high stability despite the inherent sensitivity of queue dynamics to both demand variability and discharge fluctuations. During the validation stage, the RMSE values for $Q(t)$ predictions remained comparable to those in the calibration period, further demonstrating that the FDQ model reliably captures the accumulation of excess vehicles under varying daily conditions.

Overall, the "calibration and validation" analysis indicates strong robustness of the FDQ calibration framework. The model consistently achieves low MAPE for FD and $\mu(t)$, as well as low RMSE for $Q(t)$, across multiple operating days. These results confirm the model's reliability and generalizability for continuous monitoring, congestion evaluation, and demand-management applications.

Table 9 Cross-day calibration performance of the FDQ model at the I-405 bottleneck.

Date	Weekday	Onset of congestion t_0	End of congestion t_3	Fundamental diagram		Throughput $\mu(t)$		Queue length $Q(t)$	
				RMSE	MAPE	RMSE	MAPE	RMSE	MAPE
07/03	Monday	12:10	17:25	65.36	3.60	106.99	5.29	5.54	34.28
07/05	Wednesday	12:10	19:20	62.18	3.92	144.56	8.24	7.33	38.79
07/07	Friday	11:40	19:10	78.22	4.83	148.52	9.07	7.59	35.12
07/10	Monday	13:15	20:10	87.42	5.40	143.82	7.80	7.60	39.71
07/11	Tuesday	13:15	19:45	86.87	5.67	139.14	7.72	6.76	31.33
07/24	Monday	12:55	19:00	75.56	4.98	111.62	5.30	6.52	37.76
07/25	Tuesday	13:10	20:00	91.90	5.70	155.37	8.75	8.31	34.58
07/26	Wednesday	13:15	20:40	82.38	5.19	139.98	9.88	6.44	31.03
07/27	Thursday	12:58	20:04	78.31	4.58	125.28	7.10	4.84	32.48
07/28	Friday	11:55	19:49	82.99	4.73	140.41	8.83	7.33	34.02

5.4 Efficiency loss and gain analysis

In this section, we analyze efficiency losses and potential gains using the LA dataset. To assess efficiency loss and gain from different perspective, we analyze how variations in net flow influence traffic congestion and system recovery and compare the results from three control strategies: demand-side change only, supply side change only, and demand-supply-coupled management. In this scenario-based analysis, the timestamps are kept fixed to isolate the effect of net-flow scaling. In operational implementations, the timestamps update dynamically in response to modified $\lambda(t)$ and $\mu(t)$.

5.4.1 Efficiency loss and gain analysis for FDQ-PW

We first start from the FDQ-PW model and with the demand-side management strategy only. The results are shown in Table 10, in which the net flow reduces due to a lower inflow rates (e.g., $\lambda(t_0)$ decreases), but the flow throughput remains unchanged (e.g., $\mu(t_0)$ and $\mu(t_2)$ are the same). We focus on analyzing the impact of different net flow reduction percentages on various traffic characteristics. As the net flow decreases from 9.3 veh/hr/ln to 6.5 veh/hr/ln (corresponding to a reduction from 0% to 30%), key traffic parameters change accordingly, reflecting varying levels

of congestion relief. As net flow decreases, the arrival and departure flow rates $\lambda(t_0)$, $\mu(t_0)$, $\mu(t_2)$ remain relatively stable but show a slight downward trend. This indicates that despite the reduction in traffic, the system still maintains a relatively stable operating conditions.

However, the maximum queue length $Q(t_2)$ significantly decreases, dropping from 53 vehicles to 37 vehicles, indicating that fewer vehicles are accumulating in queues as the inflow rate is reduced. Similarly, the density $k(t_2)$ decreases from 95 veh/km/ln to 76 veh/km/ln, showing that congestion is gradually alleviated, and road utilization decreases. More notably, the changes in backward wave speed (w_b) and total delay (W) demonstrate the efficiency gain due to flow reduction. As net flow decreases, w_b steadily increases from 12.09 km/hr to 17.27 km/hr, indicating that congestion dissipates more quickly. Simultaneously, the total delay W significantly decreases, from 232.95 hours to 163.06 hours, almost cutting in half. This suggests that reducing net flow effectively shortens delay time and improves roadway recovery capability.

Furthermore, the average delay drops from 0.15 veh/min to 0.11 veh/min, indicating that the congestion level decreases significantly, leading to shorter waiting times for individual vehicles. As the traffic flow reduction increases, the system experiences fewer queued vehicles, resulting in a faster dissipation of congestion and improved overall traffic efficiency. This reduction suggests that vehicles can pass through the system with fewer interruptions, enhancing roadway performance and reducing travel delays.

Table 10 Demand-side change using FDQ-PW modeling in LA case.

Net flow	Reduce %	$\lambda(t_0)$	$\mu(t_0)$	w_b	$\lambda(t_2)$	$\mu(t_2)$	$k(t_2)$	$Q(t_2)$	W	$\bar{w}$
9.3	0%	1,688	1,679	12.09	916	907	95	53	232.95	0.15
8.3	10%	1,687	1,679	13.43	915	907	89	48	209.65	0.14
7.4	20%	1,686	1,679	15.11	914	907	82	43	186.36	0.12
6.5	30%	1,685	1,679	17.27	913	907	76	37	163.06	0.11

This supply-side management strategy results are shown in Table 11, which examines how different levels of outflow rate affects key traffic parameters. In other words, the net flow reduces due to a higher throughput (e.g., $\mu(t_0)$ and $\mu(t_2)$ increases), but the inflow rates (e.g., $\lambda(t_0)$) remains unchanged. As outflow rate increases (i.e., outflow rate degradation gets improved) and the net flow decreases from 9.3 veh/hr/ln to 6.5 veh/hr/ln with reduction percentages ranging from 0% to 30%, various congestion indicators respond accordingly, reflecting the impact of supply-side changes on overall traffic performance. The arrival and departure flow rates exhibit distinct trends. The arrival rate at t_0 remains constant at 1,688 veh/h, indicating a stable incoming demand. Meanwhile, the departure rate at t_0 slightly increases from 1,679 veh/hr/ln to 1,682 veh/hr/ln, suggesting an improvement in network efficiency as congestion is mitigated. At t_2, the arrival rate $\lambda(t_2)$ remains fixed at 916 veh/h, while the throughput $\mu(t_2)$ rises steadily from 907 veh/hr/ln to 909 veh/hr/ln as the reduction percentage increases, further confirming that traffic conditions improve over time.

The queue length and density metrics also highlight congestion relief. The maximum queue length $Q(t_2)$ decreases significantly from 53 vehicles to 37 vehicles, meaning fewer vehicles are

stuck in traffic as net flow decreases. Similarly, density $k(t_2)$ drops from 95 veh/km/ln to 76 veh/km/ln, reinforcing the trend of reduced congestion.

Another critical factor is the backward wave speed w_b, which reflects how quickly congestion dissipates along the roadway. As the net flow reduction increases, w_b rises from 12.09 km/hr to 17.21 km/hr, indicating that congestion is cleared at a faster rate under improved throughput conditions. This improvement directly contributes to a substantial reduction in total delay W, which declines from 232.95 hours to 163.06 hours as the net flow is reduced by 30%. Although the reduction is moderate, the improvement in overall traffic recovery is significant.

The average delay $\bar{w}$ further illustrates the efficiency gains. As congestion becomes less severe, the average delay experienced by each vehicle decreases from 0.15 min/veh to 0.11 min/veh. This reduction indicates that individual vehicles spend less time in the queue due to shorter queue lengths and smoother discharge conditions. Overall, the supply-side improvement leads to a more efficient dissipation of congestion and enhanced operational performance.

Table 11 Supply-side change using FDQ-PW modeling in LA case.

Net flow	Reduce %	$\lambda(t_0)$	$\mu(t_0)$	w_b	$\lambda(t_2)$	$\mu(t_2)$	$k(t_2)$	$Q(t_2)$	W	$\bar{w}$
9.3	0%	1,688	1,679	12.09	916	907	95	53	232.95	0.15
8.3	10%	1,688	1,680	13.42	916	908	89	48	209.65	0.14
7.4	20%	1,688	1,681	15.08	916	908	82	43	186.36	0.12
6.5	30%	1,688	1,682	17.21	916	909	76	37	163.06	0.11

Table 12 examines the demand-supply-coupled management scenario, in which both the arrival flow $\lambda(t)$ and the realized throughput $\mu(t)$ co-evolve as the net flow progressively decreases. Unlike the supply-side-only setting, this coupled case exhibits a clear increase in both $\lambda(t_2)$ and $\mu(t_2)$ across all reduction levels. The arrival rate at t_2, increases from 916 veh/hr/ln (0 percent reduction) to 1,145 veh/hr/ln (30 percent reduction), while the departure rate grows from 907 veh/hr/ln to 1,138 veh/hr/ln over the same range. This pattern indicates that as congestion is alleviated, the system is able to process a greater number of vehicles, reflecting improved operational efficiency.

Queueing and density metrics further reinforce this trend. The maximum queue length $Q(t_2)$ declines steadily from 53 to 37 vehicles, and the density $k(t_2)$ correspondingly decreases from 95 to 76 veh/km/ln, demonstrating that congestion is increasingly mitigated as the combined demand-supply adjustments take effect.

A noteworthy distinction in this coupled scenario is that the backward wave speed w_b remains constant at 12.09 km/h across all reduction levels. This contrasts with the supply-side-only case, where w_b increased as congestion dissipated. The fixed w_b suggests that although congestion is reduced, the propagation characteristics of the congestion wave remain stable under joint demand-supply coordination.

Both the total delay W and the average delay per vehicle $\bar{w}$ exhibit a consistent downward trend, with W decreasing from 232.95 to 163.06 hours and $\bar{w}$ decreasing from 0.15 to 0.10 min/veh.

These reductions confirm that a coupled demand-supply strategy improves traffic conditions not merely by lowering congestion levels but also by enabling more efficient utilization of roadway capacity. Overall, the results highlight that integrated demand-supply management can be more effective than isolated interventions by jointly improving both congestion indicators and vehicle processing capability.

Table 12 Demand-supply coupled change using FDQ-PW modeling in LA case.

Net flow	Reduce %	$\lambda(t_0)$	$\mu(t_0)$	w_b	$\lambda(t_2)$	$\mu(t_2)$	$k(t_2)$	$Q(t_2)$	W	$\bar{w}$
9.3	0%	1,688	1,679	12.09	916	907	95	53	232.95	0.15
8.3	10%	1,687	1,679	12.09	992	984	89	48	209.65	0.13
7.4	20%	1,686	1,679	12.09	1,068	1,061	82	43	186.36	0.12
6.5	30%	1,685	1,679	12.09	1,145	1,138	76	37	163.06	0.10

Table 13 compares how key traffic parameters respond to 10-30 percent net flow increases under demand-side, supply-side, and demand-supply coupled strategies. The maximal arrival rate $\lambda(t_1)$ shows negligible changes under demand-only and no change under supply-only adjustments, whereas the coupled strategy produces substantial increases up to 25 percent, indicating improved inflow accommodation. A similar pattern appears for minimal throughput $\mu(t_2)$: demand-only yields no improvement and supply-only yields marginal gains, while the coupled strategy sharply increases throughput to 25.47 percent at 30 percent net flow increase.

Congestion indicators, including maximum queue length $Q(t_2)$, total delay W, and average delay $\bar{w}$, grow proportionally with net flow increases across all strategies. While $Q(t_2)$ and W increase uniformly regardless of the strategy, average delay rises more steeply under the coupled approach, reflecting greater system loading due to higher inflow and throughput.

Overall, only the demand-supply coupled strategy meaningfully enhances both inflow and throughput capacity, whereas demand-only and supply-only adjustments offer minimal improvements under increasing demand.

Table 13 Analysis of parameter changes under different net flow increase with FDQ-PW in LA case.

Parameters	Net flow increase	10%	20%	30%
Maximum inflow rate: $\lambda(t_1)$	Demand-side only	0.11%	0.22%	0.33%
	Supply-side only	0%	0%	0%
	Demand-supply coupled	**8.30%**	**16.59%**	**25.00%**
Minimal throughput: $\mu(t_2)$	Demand-side only	0%	0%	0%
	Supply-side only	0.10%	0.20%	0.31%
	Demand-supply coupled	**8.49%**	**16.98%**	**25.47%**
Maximum queue length: $Q(t_2)$	Demand-side only	9.43%	18.87%	30.19%
	Supply-side only	9.43%	18.87%	30.19%
	Demand-supply coupled	9.43%	18.87%	30.19%

Total delay: W	Demand-side only	10%	20%	30%
	Supply-side only	10%	20%	30%
	Demand-supply coupled	10%	20%	30%
Average delay: $\overline{w}$	Demand-side only	9.80%	19.61%	29.41%
	Supply-side only	9.80%	19.61%	29.41%
	Demand-supply coupled	**12.42%**	**24.18%**	**35.29%**

5.4.2 Efficiency loss and gain for FDQ-PN

Table 14 presents how demand-induced net flow reductions influence traffic performance under the FDQ-PN model. The polynomial degradation coefficient γ decreases from 1.71 at 0 percent reduction to 1.20 at 30 percent, indicating weaker sensitivity of throughput to congestion and thus more stable operational conditions. Correspondingly, both the arrival flow $\lambda(t_1)$ and the realized throughput $\mu(t_1)$ increase (from 1,359 to 1,453 veh/h and from 1,342 to 1,443 veh/hr/ln, respectively), reflecting enhanced flow processing capability. Congestion indicators also improve: the maximum queue length Q_{max} drops markedly from 13.74 to 9.62 vehicles, and total waiting time W is reduced from 268 to 188 hours. The congestion duration P remains constant at 8.66 hours. Taken together, increasing reduction percentages lead to higher throughput, shorter queues, and lower system delay, demonstrating improved operational efficiency under the FDQ-PN formulation.

Table 14 Demand-supply coupled change using FDQ-PN modeling in LA case.

Reduce%	γ	$\lambda(t_1)$	$\mu(t_1)$	P	Q_{max}	W
0%	1.71	1,359	1,342	8.66	13.74	268
10%	1.54	1,388	1,376	8.66	12.37	241
20%	1.37	1,421	1,409	8.66	10.99	214
30%	1.20	1,453	1,443	8.66	9.62	188

Table 15 shows that as the controlled net flow increases from 10 to 30 percent, both the arrival flow $\lambda(t_1)$ and throughput $\mu(t_1)$ rise proportionally, reaching 6.92 and 7.53 percent increases, respectively. Congestion indicators exhibit a similar linear trend: the maximum queue length Q_{max} grows from 9.97 to 29.99 percent, and total waiting time W increases from 10.07 to 29.85 percent. Notably, $\mu(t_1)$ increases more than $\lambda(t_1)$, reflecting the FDQ-PN model's sensitivity of bottleneck throughput to demand changes. Overall, higher net flow leads to coordinated increases in inflow, outflow, and congestion levels.

Table 15 Analysis of parameter changes under different net flow increase with FDQ-PN in LA case.

Parameters	Net flow increase	10%	20%	30%
$\lambda(t_1)$	Demand-supply coupled	2.13%	4.56%	6.92%
$\mu(t_1)$	Demand-supply coupled	2.53%	4.99%	7.53%

Q_{max}	Demand-supply coupled	9.97%	20.01%	29.99%
W	Demand-supply coupled	10.07%	20.15%	29.85%

5.5 Sensitive analysis of the shape parameter γ

The shape parameter γ controls the curvature of the polynomial net-flow profile and therefore governs how sharply the inflow-outflow imbalance develops within the congestion window. A larger γ yields a steeper imbalance pattern, producing sharper queue build-up and more concentrated delay, whereas a smaller γ generates a flatter and more diffused congestion pattern. Since inflow $\lambda(t)$, outflow $\mu(t)$ and queue length $Q(t)$ are analytically derived from the net-flow function, their estimation accuracy is directly influenced by the calibration of γ.

Following the sensitivity-testing approach of Cheng et al. (2024), we perturb the calibrated baseline value γ^* by $\pm 20\%$ and $\pm 40\%$ while holding all other parameters (Cap, θ, φ) fixed. For each scenario, $\lambda(t)$, $\mu(t)$ and $Q(t)$ are recomputed using the closed-form relations in Section 4.

Table 16 reports the resulting RMSE and MAPE values for inflow, outflow and queue length. The results exhibit a clear pattern. When $\gamma < \gamma^*$, the net-flow curve becomes flatter, producing more diffused congestion and smaller queue peak, which yields moderately reduced errors (e.g., lower RMSE in the $0.8\gamma^*$ scenario relative to $0.6\gamma^*$). When $\gamma > \gamma^*$, the steeper imbalance intensifies peak formation and increases the model's sensitivity to small deviations, leading to noticeably larger errors, particularly from the $1.2\gamma^*$ to the $1.4\gamma^*$ scenario. The baseline $\gamma = \gamma^*$ consistently achieves the lowest RMSE across all variables, confirming that the calibrated shape parameter most accurately represents the true congestion dynamics.

Table 16 Sensitivity of the shape parameter γ on inflow, outflow, and queue estimation performance.

Scenario	Value	Inflow rate (RMSE)	Outflow rate (RMSE)	Queue length (RMSE)	Inflow rate (MAPE)	Outflow rate (MAPE)	Queue length (MAPE)
Decrease ($0.6\gamma^*$) γ_1	1.03	223.0	215.8	16.3	17.0	16.2	43.1
Decrease ($0.8\gamma^*$) γ_2	1.37	160.3	157.2	10.6	11.6	11.3	33.1
Baseline ($1.0\gamma^*$) γ_0	1.71	153.6	157.4	7.8	10.0	10.1	33.3
Increase ($1.2\gamma^*$) γ_3	2.05	208.4	216.3	10.7	14.3	14.8	44.5
Increase ($1.4\gamma^*$) γ_4	2.39	292.0	301.2	16.5	21.5	22.0	61.0

5.6 Implementation feasibility and simulation integration

Although the preceding subsections focused on the analytical evaluation of the proposed demand- and supply-side strategies, the FDQ framework is designed to be operable within simulation environments and transferable to field-replicable settings. We emphasize at the outset that the integrations with simulation platforms discussed in this subsection are intended to

demonstrate theoretical and interface-level feasibility, rather than to report completed simulation-based experiments. The objective of this section is to clarify implementation pathways, identify sources of uncertainty, and delineate appropriate platforms for future validation.

The calibrated FDQ parameters inevitably exhibit an error margin of approximately 10-15 percent. This uncertainty arises from three main sources. First, loop-detector measurements of flow, occupancy, and density are subject to noise and lane-aggregation bias, particularly under highly turbulence congested conditions. Second, multi-lane interactions such as merging, weaving, and cooperative maneuvers cannot be fully captured within a single-bottleneck analytical representation. Third, the adoption of a unified fundamental-diagram formulation over an extended congestion period may overlook short-term heterogeneity in traffic composition. These limitations motivate the need of validating FDQ within simulation environment that explicitly represent vehicle interactions and spatiotemporal variability.

Despite its analytical formulation, the FDQ model is theoretically implementable in commonly used microscopic simulators platforms such as VISSIM or SUMO at the interface level. Specifically, the FDQ-generated dynamic arrival profile $\lambda(t)$ can be specified as a time-dependent demand input, while the time-varying throughput function $\mu(t)$ can be mapped to a dynamic capacity constraint at the bottleneck link or connector. Once such mappings are defined, microscopic simulators can reproduce congestion breakdown and recovery through their embedded car-following, lane-changing, and gap-acceptance models, thereby generating flow variability that extends beyond the analytical structure of FDQ. These mappings clarify the feasibility of embedding FDQ outputs into microsimulation platforms; however, they do not constitute a full-scale microsimulation-based validation, which is beyond the scope of the present study.

For mesoscopic settings, integration is conceptually more direct. Within a CTM, the FDQ-derived throughput $\mu(t)$ replaces the downstream receiving flow of the bottleneck cell, while the FDQ arrival trajectory defines the upstream sending flow. Because CTM formulations are inherently based on the fundamental-diagram supply-demand relationships, this substitution preserves theoretical consistency while enabling congestion propagation and dissipation to be evaluated over multiple cells. In this context, FDQ-based interventions can be assessed at the corridor level, allowing the analyst to examine congestion duration, delay accumulation, dissipation dynamics, and improvements in discharge stability.

Taken together, these microscopic and mesoscopic integrations demonstrate that the FDQ framework is operationally compatible with widely used simulation tools at the interface level. The model outputs, including dynamic arrival rates, time-varying throughput, and queue evolution, can serve as direct inputs to simulation platforms, providing a practical pathway for future validation and evaluation of congestion-management strategies under realistic traffic conditions.

The overall implementation workflow is summarized in Fig. 19, which illustrates the progression from data input and parameter calibration to generation of dynamic arrival and throughput functions, followed by their integration into microscopic simulators (e.g., VISSIM or SUMO) at a conceptual level and mesoscopic models such as CTM. This workflow demonstrates how the FDQ framework transitions from an analytical formulation to a simulation-ready structure, thereby addressing key considerations related to practical applicability and operational feasibility. To further strengthen the empirical relevance of the proposed framework, simulation-based validation is identified as an important direction for future research.

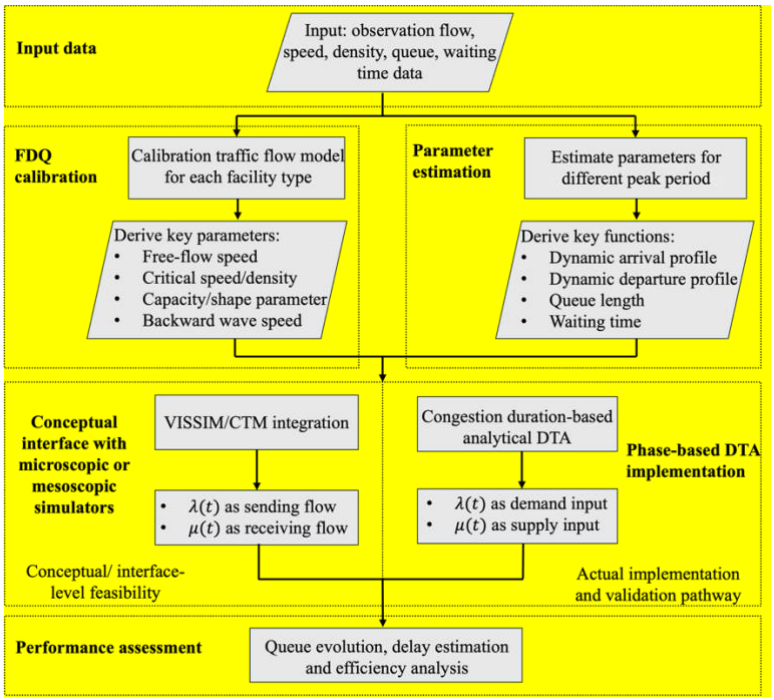


Figure 19 Workflow of the FDQ model and its integration into simulation platforms.

It is important to acknowledge the current limitations of the implementation discussion presented above. The integration pathways with microscopic simulators are conceptual and require empirical validation in future work. A fundamental challenge lies in the methodological mismatch between FD-consistent analytical models and the behavioral paradigms of most second-by-second microscopic simulators. Platforms such as VISSIM or SUMO generate traffic dynamics through local behavioral rules that may not inherently preserve the aggregate fundamental-diagram relationships observed in field data. This creates a trade-off between "simulation reality" (what the simulator produces based on its own behavioral rules) and "observed reality" (what field data shows in terms of FD-consistent throughput dynamics).

For meaningful validation and operational deployment, we propose a specific implementation pathway through phase-based dynamic traffic assignment (DTA) frameworks rather than conventional second-by-second microscopic simulators. This pathway leverages the restricted master problem formulation commonly used in DTA optimization (Lu et al., 2009), where our FDQ model can provide two critical components:

(i) Time-dependent link performance functions: The FDQ-derived waiting times and throughput dynamics serve as inputs to the restricted master nonlinear programming problem, replacing traditional BPR-type or constant discharge rate functions. This enables the DTA optimization to minimize system-level objectives (e.g., equivalent gap functions as formulated in Lu et al., 2009, total system delay) while respecting FD-consistent bottleneck behavior.

(ii) Time-dependent loading matrices: The FDQ model generates dynamic throughput profiles that can be directly incorporated into the time-dependent traffic loading process, enabling the DTA to load flows onto the network in a manner consistent with observed FD relationships rather than purely simulation-generated flow patterns.

This approach establishes a co-simulation and co-optimization framework where the simulation platform employs a weighting mechanism between its internal behavioral dynamics and the FD-consistent analytical properties provided by FDQ. The weighting function would be calibrated based on the confidence level in field-derived FD parameters versus simulation-generated flow patterns. For instance, if the FDQ model demonstrates superior approximation to observed reality (validated through field data), it receives higher weight in the restricted master problem formulation. This hybrid approach preserves both the behavioral realism of simulation and the analytical rigor of FD-consistent performance functions.

One concrete implementation platform is OpenDTA (https://github.com/jdlph/OpenDTA), which supports this type of phase-based, restricted master problem formulation with customizable link performance functions. In this framework, FDQ provides aggregated, analytically derived properties suitable for high-level co-simulation and co-optimization, rather than requiring second-by-second microscopic resolution. This represents a more natural integration pathway for FDQ, as it operates at the same temporal aggregation level (phase-based rather than second-by-second) and explicitly couples demand-supply interactions through the restricted master problem structure.

6 Discussion

6.1 Interrupted flow

Although the FDQ model is developed for freeway bottlenecks, its structure naturally extends to interrupted-flow environments such as signalized arterials, where periodic supply fluctuations are intrinsic to traffic operations. At a signalized stop-line, the effective throughput capacity alternates between green and red phases, and this cyclic switching directly corresponds to time-dependent variations in $\mu(t)$. Two adaptations allow the FDQ framework to remain physically consistent under such conditions.

(1) Signal-cycle-driven $\mu(t)$

During green phases, downstream supply is positive and is well-approximated by a green-phase throughput $\mu(t) = Cap_g$, which reflects the saturation flow rate adjusted for green duration and start-up lost time. During red phases, $\mu(t)$ drops to zero or to a minimal leakage flow $\mu(t) = Cap_r \approx 0$. This results in a piecewise time-varying capacity profile, as follows:

$$\mu(t) = \begin{cases} Cap_g, t\epsilon\ green\ phase \\ Cap_r, t\epsilon\ red\ phase \end{cases} \tag{40}$$

which fits naturally within the FDQ formulation that already accommodates time-varying throughput functions.

(2) Queue spillback as a dynamic capacity constraint

When the upstream queue grows sufficiently to reach the detector or stop-line, downstream storage becomes binding and the effective throughput must be reduced. FDQ captures this phenomenon by applying a spillback constraint:

$$\mu(t) = \min\{\mu(t), \mu_{spillback}\} \tag{41}$$

where spillback reduces the effective throughput below the nominal green-phase capacity. This aligns with observed arterial behavior where oversaturation, blocking back, and residual queues

restrict outflow even during the green phase.

Beyond the physical capacity updates, the interpretation of $\lambda(t)$ and $\mu(t)$ in arterial contexts also adapts naturally. The arrival profile $\lambda(t)$ represents signal-generated vehicle platoons and thus exhibits cycle-to-cycle variability, whereas $\mu(t)$ reflects downstream discharge governed by signal timing plans, turn-bay utilization, and interactions with spillback. Together, these mechanisms enable the FDQ model to capture both arrival-driven congestion formation and signal-driven discharge modulation. Consequently, FDQ serves as a compact analytical complement to mesoscopic and microscopic arterial simulation models, particularly for studying delay patterns and oversaturated operations without requiring full network-level signal coordination.

6.2 Connected and Automated Vehicles

Recent studies have shown that CAVs can mitigate traffic turbulence and reduce the magnitude of capacity drop through cooperative merging and smoother inflow patterns (Talebpour and Mahmassani, 2016; Shi and Li, 2021; Levin and Boyles, 2016; Li et al., 2025). The FDQ framework is inherently compatible with these mechanisms.

First, the smoothing effect that CAVs exert on the arrival flow can be directly reflected as a reshaping of the upstream arrival rate $\lambda(t)$. Through coordinated platooning, speed harmonization, and synchronized acceleration, CAVs can create a more uniform and less oscillatory inflow upstream of the bottleneck. This characteristic aligns closely with the $\lambda(t)$ management strategies examined in Section 5, enabling the FDQ framework to quantify how CAV-enabled arrival coordination improves flow throughput, reduces queuing, and enhances overall system stability.

Second, the stabilization of bottleneck throughput induced by CAVs can be interpreted as a "flattening" of the $\mu(t)$ profile. By promoting smoother merging, reducing speed variability, and mitigating turbulence at the bottleneck, CAVs can alleviate the classical capacity drop phenomenon. Within the FDQ framework, these effects manifest as: (i) a reduction in short-term fluctuations of $\mu(t)$; (ii) an increase in the average $\mu(t)$ level toward its pre-capacity-drop value; (iii) a higher dissipation rate of residual queues.

These changes are fully consistent with empirical findings that the magnitude of capacity drop decreases substantially when bottleneck turbulence is suppressed. In the FDQ model, this improvement is captured analytically through the θ-$Q(t)$ relationship, which links queue length to the effective throughput.

Taken together, these connections indicate that FDQ not only captures the improvements in arrival flow and discharge stability brought by CAVs, but also provides a structured, FD-consistent analytical tool for evaluating CAV-enabled congestion mitigation strategies. Furthermore, the extensibility of the FDQ framework suggests strong potential for deeper integration with cooperative merging control, speed harmonization (V2I/V2X), and reservation-based merging, offering a foundation for developing physics-consistent, CAV-enhanced bottleneck management frameworks.

6.3 Practical operationalization of demand-supply coupling

The FDQ framework provides a natural platform for operationalizing demand–supply coupling in real traffic environments. Operationalization requires three complementary components. First, the data requirements are modest, relying primarily on real-time measurements of flow, density, and bottleneck occupancy from loop detectors or probe-vehicle data. These inputs support continuous estimation of $\lambda(t)$ and $\mu(t)$ and allow the FDQ model to update the congestion

state without full spatial reconstruction.

Second, several existing traffic control mechanisms can serve as the actuation layer for demand-supply coupling. Ramp metering and variable speed limit (VSL) systems provide direct levers to modify the arrival flow $\lambda(t)$, whereas dynamic bottleneck control, including cooperative merging, lane-use control, and reduced-speed zones, affects the realized throughput $\mu(t)$. Thus, the FDQ-based coupling aligns well with modern freeway management architectures.

Third, the FDQ formulation can be integrated into established ATDM platforms by supplying the time-varying demand and supply trajectories as inputs to predictive or rule-based controllers. ATDM systems already incorporate modules for queue override, congestion detection, and multi-objective optimization, enabling real-time adjustments to inflow and throughput rates. Embedding FDQ-generated $\lambda(t)$ and $\mu(t)$ into ATDM logic creates a unified demand-supply coordination framework capable of mitigating breakdown, shortening congestion duration, and reducing delay.

This discussion highlights that the FDQ framework is not only analytically tractable but also directly compatible with current freeway management infrastructure, supporting practical deployment at both isolated bottlenecks and corridor-wide ATDM installations.

6.4 Limitations

Several simplifying assumptions were adopted in this study to maintain analytical tractability and consistency with fundamental-diagram-based representations. While these assumptions enable closed-form expressions for queue length, cumulative flows, waiting time, and delay, they also introduce key limitations.

First, the FDQ framework relies on a linear mapping between queue length and excess density. This assumption is most appropriate for homogeneous bottlenecks with relatively constant backward wave speeds and uniform congestion propagation. In real-world environments characterized by lane-changing turbulence, multi-lane heterogeneity, variable merging intensity, or non-stationary shockwave patterns, the spatial queue profile may deviate from linearity and the effective density-queue mapping may differ by lane. Nonlinear density-queue mappings or multi-lane kinematic descriptions may be required to represent such heterogeneous congestion formation more accurately.

Second, the linear dependence of throughput $\mu(t)$ on queue length $Q(t)$ reflects the congested branch of the triangular fundamental diagram. Although this preserves FD-consistent monotonicity, it may not fully capture discharge variability due to turbulence, cooperative or forced merging, or weather-induced fluctuations. The quadratic FD extension in Section 4.1.4 partially relaxes this constraint by introducing smooth nonlinear degradation, but additional refinements, such as lane-specific calibration, multi-regime throughput models, or stochastic representations of discharge may be necessary to reflect real-world bottleneck capacity variability.

Third, the FDQ framework assumes that congestion forms at a single stationary bottleneck location with well-defined onset and dissipation times. While appropriate for recurrent bottlenecks, this assumption may not hold under intermittent downstream spillback, interacting bottlenecks, time-varying control strategies, or signal-driven capacity fluctuations. In such settings, spatial queue propagation, time-varying bottleneck locations, or corridor-level capacity constraints may need to be incorporated to generalize the FDQ framework.

Finally, the FDQ framework differs fundamentally from second-order macroscopic traffic models such as the Payne-Whitham formulation. Second-order models explicitly represent

momentum and speed evolution through acceleration, relaxation, anticipation, and pressure terms, enabling rich spatiotemporal wave dynamics that exceed the representational scope of first-order FD-based formulations. However, classical second-order models, particularly Payne-Whitham, exhibit well-documented drawbacks (Daganzo, 1995), including violations of anisotropy, negative speeds, unrealistically fast backward wave propagation, and physically inconsistent overtaking behavior. These limitations motivated improved second-order formulations such as the Aw-Rascle-Zhang (ARZ) model, though such models require full PDE-based link-level dynamics that differ substantially from the bottleneck-focused, mesoscopic queue representation used in FDQ. The FDQ framework should therefore be viewed as complementary rather than a substitute for second-order models: FDQ provides an analytically tractable, FD-consistent representation of bottleneck efficiency dynamics, whereas second-order models remain essential for capturing full spatiotemporal traffic evolution across links.

7 Conclusions and Future Directions

This paper extends the traditional fluid queue model by incorporating a stationary density–flow relationship observed empirically at key bottlenecks, referred to as FDQ, to better describe the outflow rate to evolve with local traffic state transitions, especially in the over-congestion regime. Our model thus captures the interplay between queue evolution and throughput dynamics, offering a hybrid perspective that integrates fluid conservation with state-dependent empirical closure. By integrating the FD with the fluid queuing approach, the FDQ model addresses the inconsistencies observed in popular queuing models like the PAQ when applied to over-congested regimes. Through the extension of PAQ with a dynamic flow throughput, the FDQ model achieves three key objectives: it derives FD-consistent polynomial performance evaluation models, accurately estimates dynamic traffic states during congestion to capture queue formation and dissipation and explore how slight arrival flow patterns adjustments can affect efficiency gain and reduce delay and maximum queue length. These capabilities enable the FDQ model to provide closed-form expressions for dynamic travel delay performance, enhancing its practical applicability for assessing single traffic bottlenecks.

The multi-resolution nature of the FDQ model offers a numerically reliable and theoretically rigorous performance function that operates effectively at both macroscopic and mesoscopic scales. This dual-scale functionality is crucial for accurately characterizing over-congestion bottlenecks and contributes to more effective time-dependent traffic state estimation and dynamic traffic flow control. Two datasets, the LA I-405 and Phoenix I-10, were used in the numerical analysis. Two FDQ models were examined: one with flow throughput in polynomial form and another in piecewise form. Additionally, two benchmark models, PW and PAQ, were employed for comparison and analyze the efficiency loss and gain.

The results demonstrate that: (1) The proposed FDQ modeling framework significantly improves traffic state estimation accuracy. The FDQ model effectively bridges the gap between the empirical observations and theoretical models in traffic flow theory. Its ability to capture complex traffic dynamics in congested networks holds promise for improving traffic planning, management, and control strategies; (2) Reducing the network flow rate does not necessarily require a reduction in the inflow rate and may even allow for an increase in the supported inflow rate. This opens new avenues for research and application in smart demand reservation systems; (3) Demand-supply coupled control offers a more significant efficiency gain compared to demand

or supply unilateral control. This demonstrates that strategically managing the network flow rate enables the system to sustain higher inflow and outflow rates, particularly during peak congestion periods.

Future research could focus on: (1) extending this model to address multiple interconnected bottlenecks and incorporating stochastic elements to account for traffic variability, further enhancing its robustness and applicability in real-world scenarios. Enhancing the model's precision by incorporating real-time traffic data, such as vehicle speed and acceleration patterns, could further refine its predictions; (2) investigating the impact of external factors, such as weather conditions and road incidents, on dynamic flow throughput and queue lengths would provide valuable insights and further improve the model's overall accuracy; (3) exploring the integration of CAVs within reservation-based traffic control frameworks, assessing their effectiveness in optimizing flow throughput and reducing congestion under oversaturated conditions. Investigating the interaction between human-driven vehicles and CAVs in mixed traffic environments would be essential to understanding system-level impacts on efficiency and safety; (4) while Section 5.6 demonstrates conceptual feasibility, actual implementation and validation remain important future work. The proposed co-simulation and co-optimization framework bridges the methodological disconnection between microscopic simulation and macroscopic FD-consistent properties. We acknowledge that this implementation pathway requires coordinated development between analytical models and DTA simulation platforms, which we identify as a critical direction for future research. The challenge is not merely technical validation, but establishing the appropriate coupling mechanisms and weighting schemes that balance multiple sources of traffic flow information

Appendices

Appendix A: Notation List

Notations	Definitions	Unit
t_0	Start time of congestion period	hr
t_1	Time with maximum arrival rate	hr
t_2	Time with maximum queue length	hr
t_3	End time of congestion period	hr
P	Congestion duration, $P = t_3 - t_0$	hr
$\lambda(t)$	Arrival rate at time t	veh/hr/ln
$\mu(t)$	Flow throughput at time t	veh/hr/ln
$Q(t)$	Virtual queue length at time t, measured in vehicles	veh/hr/ln
$w(t)$	Waiting time at time t	hr
$tt(t)$	Travel time at time t	hr
$\pi(t)$	Net flow at time t	veh/hr/ln
$A(t)$	Cumulative arrival curve at time t	veh/hr/ln
$D(t)$	Cumulative departure curve at time t	veh/hr/ln
γ	Shape parameter for the inflow rate $\lambda(t)$ and net flow function $\pi(t)$	NA
v_f, t_f	Free flow speed and free flow travel time	km/hr
v_μ	Congestion speed	km/hr
k_c	Critical density	veh/km/ln
Cap	Bottleneck capacity after capacity drop	veh/hr/ln
w_b	Backward wave speed	km/hr
W	Total waiting time during the whole congestion period	hr
$\bar{w}$	Average waiting time for each vehicle	hr/veh

Appendix B: Some Derivations

1. Derivation of time-dependent queue length $Q(t)$ in Eq. (17).

$$Q(t) = \int_{t_0}^{t} \pi(\tau) d\tau = \int_{t_0}^{t} \gamma(\tau - t_0)(t_2 - \tau) d\tau \tag{42}$$

We can apply substitution method, letting $u = t - t_0$, which gives us $du = dt$.

Furthermore, when $t = t_0$, we have $u = 0$, and when $t = t$, we have $u = t - t_0$

Therefore, we have:

$$\begin{aligned} Q(t) = \int_{t_0}^{t} \gamma(\tau - t_0)(t_2 - \tau) d\tau &= \int_{0}^{u} \gamma u[t_2 - (u + t_0)]\, du \\ &= \gamma \int_{0}^{u} u[(t_2 - t_0) - u)]\, du = \gamma \int_{0}^{u} u[(t_2 - t_0)] du - \gamma \int_{0}^{u} u^2 du \\ &= \gamma(t_2 - t_0) \int_{0}^{u} u du - \gamma \int_{0}^{u} u^2 du = \gamma(t_2 - t_0) \frac{1}{2} u^2 - \gamma \frac{1}{3} u^3 \end{aligned} \tag{43}$$

Bringing $u = t - t_0$ back into the above equation, we have:

$$\begin{aligned} Q(t) = \gamma(t_2 - t_0) \frac{1}{2} u^2 - \gamma \frac{1}{3} u^3 &= \gamma(t_2 - t_0) \frac{1}{2} (t - t_0)^2 - \gamma \frac{1}{3} (t - t_0)^3 \\ &= \gamma(t - t_0)^2 [\frac{(t_2 - t_0)}{2} - \frac{(t - t_0)}{3}] \end{aligned} \tag{44}$$

The derivation of Eq. (17) is thus complete.

2. Derivation of cumulative departure flow profile $D(t)$ in Eq. (22)

$$\begin{aligned} D(t) = \int_{t_0}^{t} \mu(\tau) d\tau = \int_{t_0}^{t} \left[Cap - \theta \cdot \frac{\gamma}{3} (\tau - t_0)^2 (t_3 - \tau) \right] d\tau = \\ = \int_{t_0}^{t} Cap d\tau - \theta \cdot \frac{\gamma}{3} \int_{t_0}^{t} [(\tau - t_0)^2 (t_3 - \tau)] d\tau \end{aligned} \tag{45}$$

The first component is easy to solve, $\int_{t_0}^{t} Cap d\tau = Cap \cdot (t - t_0)$

To compute the second component, we will apply substitution method, letting $u = \tau - t_0$, which gives us $du = d\tau$.

Furthermore, when $\tau = t_0$, we have $u = 0$, and when $\tau = t$, we have $u = t - t_0$

Therefore, we have:

$$-\theta \cdot \frac{\gamma}{3}\int_{t_0}^{t}[(\tau - t_0)^2(t_3 - \tau)]d\tau = -\theta \cdot \frac{\gamma}{3}\int_{0}^{t-t_0} u^2[(t_3 - t_0) - u]du \tag{46}$$

$$= -\theta \cdot \frac{\gamma}{3}\int_{0}^{t-t_0}[u^2(t_3 - t_0) - u^3]du$$

$$= -\theta \cdot \frac{\gamma}{3} \cdot [\frac{u^3}{3}(t_3 - t_0) - \frac{u^4}{4}]|_0^{t-t_0}$$

$$= -\theta \cdot \frac{\gamma}{3} \cdot [\frac{(t - t_0)^3}{3}(t_3 - t_0) - \frac{(t - t_0)^4}{4}]$$

$$= -\theta \cdot \frac{\gamma}{3} \cdot (t - t_0)^3[\frac{t_3 - t_0}{3} - \frac{t - t_0}{4}]$$

Combining the results from the first and second components, we have:

$$D(t) = Cap \cdot (t - t_0) - \theta \cdot \frac{\gamma}{3} \cdot (t - t_0)^3 \cdot (\frac{t_3 - t_0}{3} - \frac{t - t_0}{4}) \tag{47}$$

The derivation of Eq. (22) is thus complete.

3. Derivation of cumulative arrival flow profile $A(t)$ in Eq. (23)

$$A(t) = \int_{t_0}^{t}\lambda(\tau)d\tau = \int_{t_0}^{t}[\mu(\tau) + \pi(\tau)]d\tau = \int_{t_0}^{t}\mu(\tau)d\tau + \int_{t_0}^{t}\pi(\tau)d\tau \tag{48}$$

The first component $\int_{t_0}^{t}\mu(\tau)d\tau$ equals to $D(t)$, and has been derived in Eq. (22).

The first component $\int_{t_0}^{t}\pi(\tau)d\tau$ equals to $Q(t)$, and has been derived in Eq. (17).

Combined, we have:

$$A(t) = D(t) + Q(t) \tag{49}$$

$$= \gamma\left(\frac{1}{3}t_0 + \frac{2}{3}t_3\right) \cdot \frac{(t - t_0)^2}{2} - \gamma(t - t_0)^2\left(\frac{t - t_0}{3} - \frac{t_0}{2}\right) + Cap$$

$$\cdot (t - t_0) - \theta \cdot \frac{\gamma}{3} \cdot (t - t_0)^3 \cdot (\frac{t_3 - t_0}{3} - \frac{t - t_0}{4})$$

The derivation of Eq. (23) is thus complete.

4. Derivation of total delay W in Eq. (28)

$$W = \int_{t_0}^{t_3}Q(t)dt = \int_{t_0}^{t_3}\frac{\gamma}{3}(t - t_0)^2(t_3 - t)dt = \frac{\gamma}{36}(t_3 - t_0)^4 \tag{50}$$

We can apply substitution method, letting $u = \frac{t-t_0}{t_3-t_0}$, which gives us $t = t_0 + u(t_3 - t_0)$, and then $dt = (t_3 - t_0)du$.

Additionally, we will have $t - t_0 = u(t_3 - t_0)$, and that:

$$t_3 - t = t_3 - [t_0 + u(t_3 - t_0)] = (1 - u)(t_3 - t_0) \tag{51}$$

Furthermore, when $t = t_0$, we have $u = 0$, and when $t = t_3$, we have $u = 1$.

Then the above equation could be rewritten as:

$$\begin{aligned} W &= \int_{t_0}^{t_3} \frac{\gamma}{3}(t - t_0)^2(t_3 - t)dt \\ &= \int_0^1 \frac{\gamma}{3}[u(t_3 - t_0)]^2 \cdot [(1 - u)(t_3 - t_0)] \cdot (t_3 - t_0)du \\ &= \frac{\gamma}{3}(t_3 - t_0)^4 \int_0^1 u^2 \cdot (1 - u)du = \frac{\gamma}{3}(t_3 - t_0)^4 \int_0^1 (u^2 - u^3)du \\ &= \frac{\gamma}{3}(t_3 - t_0)^4 \left(\frac{1}{3} - \frac{1}{4}\right) = \frac{\gamma}{36}(t_3 - t_0)^4 \end{aligned} \tag{52}$$

The derivation of Eq. (28) is thus complete.

Appendix C: Detailed Formulation and Efficiency Analysis of the FDQ-PW Framework

In this subsection, we adopt a piecewise linear form to represent the arrival and departure rates. The rationale behind this choice is that compared to polynomial functions, piecewise functions are significantly simpler and, therefore, easier to calibrate. This makes them potentially particularly useful in real-world engineering applications, yet still with reasonable modeling accuracy.

As illustrated in Fig. 20, following the FDQ-PW modeling approach, most variables are expressed in linear and piecewise forms. This includes the arrival and departure rates in Fig. 20(a), the queue profile in Fig. 20(b), and the network flow in Fig. 20(d). The only exception is the cumulative arrival and departure flow profiles in Fig. 20(c), which is in quadratic form because it is the sum of inbound and outbound flows and is therefore one degree higher. Specifically, in Fig. 20(a), the blue and red straight lines represent the time-dependent arrival rate $\lambda(t)$ and throughput rate $\mu(t)$, respectively, both expressed in piecewise linear form. The queue exhibits a linear change trend as depicted in Fig. 20(b), initially forming and then dissipating. Fig. 20(c) displays the cumulative arrival and cumulative departure curves, while Fig. 20(d) shows the net flow rate curve, where the net flow rate is positive during the period $[t_0, t_2]$ and negative during the period $[t_2, t_3]$.

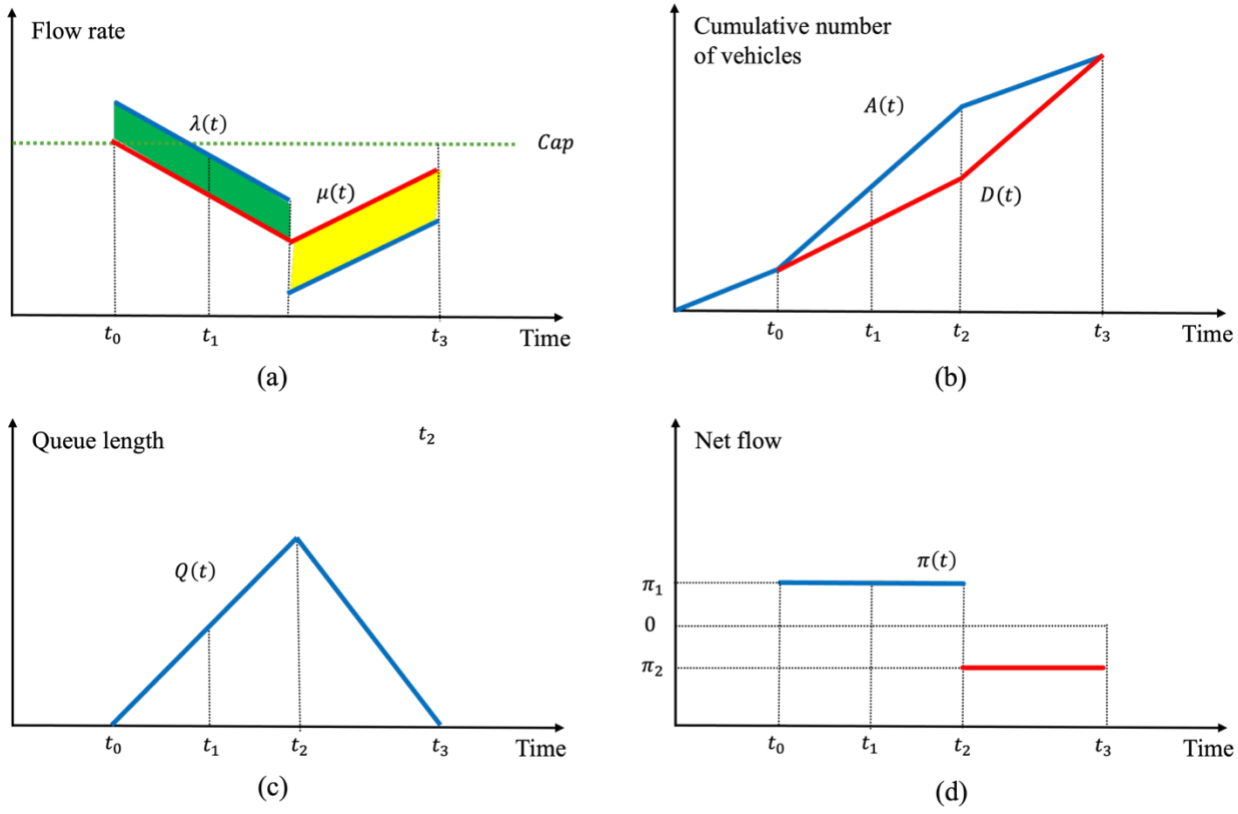


Figure 20 Illustration of FDQ-PW with piecewise queue models: (a) arrival and departure rate; (b) cumulative arrival and departure rate; (c) queue profile; (d) net flow profile.

Similar to the derivation process outlined above for the FDQ-PN model, we begin by analyzing the net flow profile $\pi(t)$ presented in Fig. 20(d). In the FDQ-PW model, since the queue grows linearly, the net flow has to be a constant. As such, the net flow can be expressed as following:

$$\pi(t) = \begin{cases} \pi_1 > 0, t\epsilon(t_0, t_2) \\ \pi_2 < 0, t\epsilon(t_2, t_3) \end{cases} \tag{53}$$

Then we could derive the piecewise queue length profile, as shown in the following:

$$Q(t) = \begin{cases} \pi_1(t - t_0), t\epsilon(t_0, t_2) \\ -\pi_2(t_3 - t), t\epsilon(t_2, t_3) \end{cases} \tag{54}$$

In Fig. 20(b), the queue length reaches its maximum at time t_2, from which we can derive the net flow π_1 and π_2:

$$\pi_1 = \frac{Q(t_2)}{t_2 - t_0} \tag{55}$$

$$\pi_2 = -\frac{Q(t_2)}{t_3 - t_2} \tag{56}$$

Based on Eq. (56), the relationship between critical time points can also be obtained, as shown in the following:

$$t_3 = t_2 - \frac{\pi_1(t_2 - t_0)}{\pi_2} \tag{57}$$

We will move onto the derivation of the flow throughput $\mu(t)$ and the arrival rate $\lambda(t)$, both in linear piecewise form. For $\mu(t)$, we have $\mu(t) = Cap - \theta \cdot Q(t)$. If we substitute the $Q(t)$ with the expression from Eq. (54), we will get:

$$\mu(t) = \begin{cases} Cap - \pi_1 \cdot \theta \cdot (t - t_0), t\epsilon(t_0, t_2) \\ Cap + \pi_2 \cdot \theta \cdot (t_3 - t), t\epsilon(t_2, t_3) \end{cases} \tag{58}$$

Additionally, since the net flow rate is the difference between the arrival flow rate and the departure flow rate, i.e., $\pi(t) = \lambda(t) - \mu(t)$, we could derive the arrival flow rate as $\mu(t) + \pi(t)$:

$$\lambda(t) = \begin{cases} Cap + \pi_1[1 - \theta(t - t_0)], t\epsilon(t_0, t_2) \\ Cap + \pi_2[1 + \theta(t_3 - t)], t\epsilon(t_2, t_3) \end{cases} \tag{59}$$

The total delay between the time t_0 and t_3 can also be calculated by integrating the queue length $Q(t)$, as shown below:

$$\begin{aligned} W = \int_{t_0}^{t_3} Q(t)dt &= \int_{t_0}^{t_2} \pi_1(t - t_0)dt - \int_{t_2}^{t_3} \pi_2(t_3 - t)dt \\ &= \frac{1}{2}[\pi_1(t_2 - t_0)^2 + \pi_2(t_3 - t_2)^2] \end{aligned} \tag{60}$$

As mentioned above, we could model piecewise linear throughput consistency with the quadratic FD. The illustration of queue system of the piecewise linear flow throughput consistent with quadratic FD is shown Fig. 21.

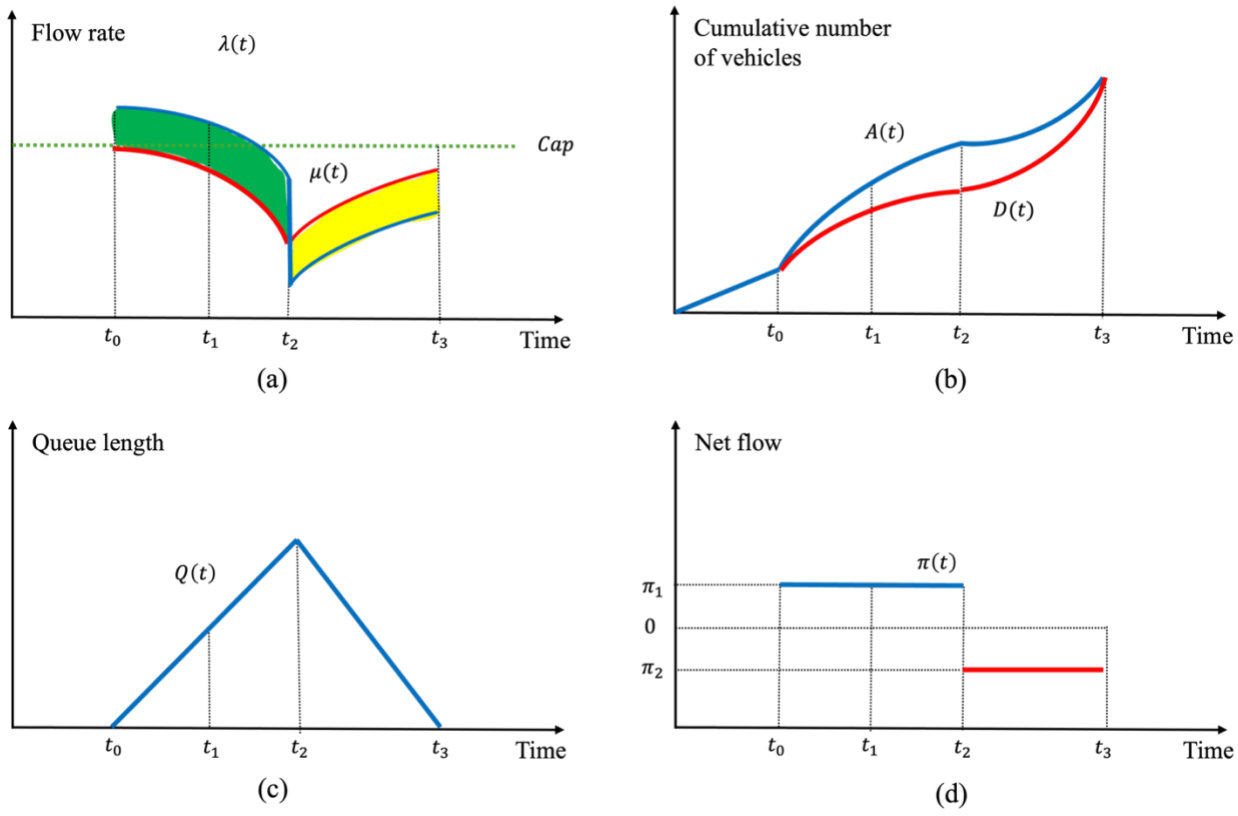


Figure 21 Illustration of the queuing system with piecewise linear flow throughput consistent with the quadratic FD.

The flow throughput of the bottleneck, as shown in the following:

$$\mu(t) = \begin{cases} Cap - \frac{Cap}{{k_j}^2} \cdot [\pi_1(t - t_0)]^2, t\epsilon(t_0, t_2) \\ Cap - \frac{Cap}{{k_j}^2} \cdot [\pi_2(t_3 - t)]^2, t\epsilon(t_2, t_3) \end{cases} \tag{61}$$

The arrival flow rate, as shown in the following:

$$\lambda(t) = \begin{cases} Cap + \pi_1 - \frac{Cap}{{k_j}^2} \cdot [\pi_1(t - t_0)]^2, t\epsilon(t_0, t_2) \\ Cap + \pi_2 - \frac{Cap}{{k_j}^2} \cdot [\pi_2(t_3 - t)]^2, t\epsilon(t_2, t_3) \end{cases} \tag{62}$$

Similar to the previous process, Fig. 22 illustrates the efficiency loss and gain for the FDQ-PW model from three perspectives. In each case, both the arrival rate (red curve) and flow throughput (blue curve) are modeled using linear piecewise functions. The objective is to quantify how modest adjustments to arrival flow patterns, representing inflow control can enhance

throughput while reducing queue length and delay. In Fig. 22(a), the inflow rate is adjusted while keeping throughput fixed, simulating strategies such as reservations or tolling. In Fig. 22(b), the inflow remains constant while throughput is modified, reflecting interventions like ramp metering or directional flow control. Fig. 22(c) presents a combined strategy, adjusting both inflow and throughput simultaneously.

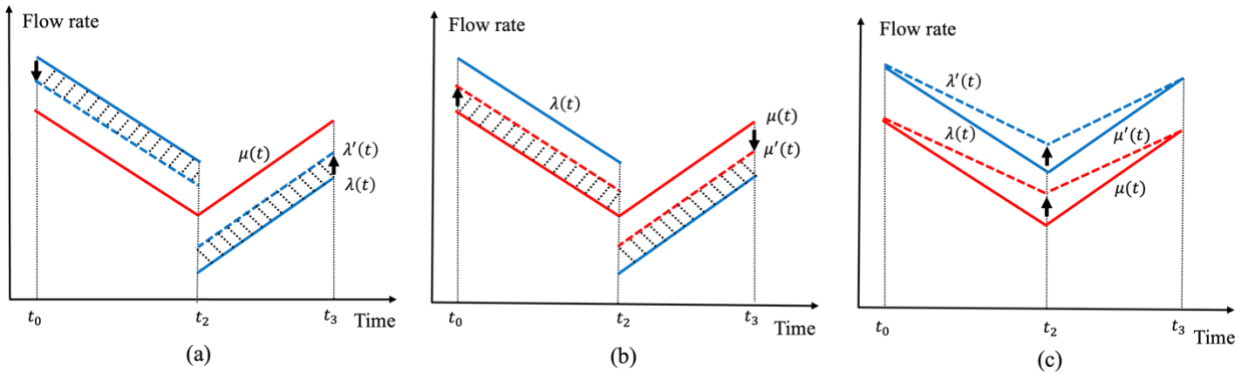


Figure 22 Illustration of different efficiency gain in the FDQ-PW: (a) demand-side change strategy; (b) supply-side change strategy; (c) demand-supply coupled change strategy.

The detailed progress to analyzing efficiency loss and gain with the FDQ-PW model is shown in Algorithm 2. Compared to Algorithm 1, which adopts a polynomial form of flow throughput, Algorithm 2 uses a piecewise form where the outflow $\mu'(t)$ is linearly reduced over time according to the net flow π' and the reduction rate θ, i.e., $\mu'(t) = Cap - \pi' \cdot \theta \cdot (t - t_0)$. Consequently, arrival rates are updated via the identity $\lambda'(t) = \mu'(t) + \pi'$, ensuring demand-supply equilibrium. Performance metrics such as queue length, delay, and density are estimated using piecewise integration and linear accumulation. Unlike FDQ-PN, FDQ-PW also calculates updated density and fundamental diagram parameters, including backward wave speed and jam density, to quantify how adjustments in flow throughput influence wave dynamics. This structure enables clearer identification of efficiency gains tied to congestion formation and dissipation under piecewise queuing conditions.

Algorithm 2: Efficiency loss and gain analysis using FDQ-PW modeling approach.

Step 1: Benchmark parameter calculation: Calibrate the initial benchmark parameters with the field data: net flow π, arrival rate at t_0, $\lambda(t_0)$, flow throughput at t_0, $\mu(t_0)$, arrival rate at t_2, $\lambda(t_2)$, flow throughput at t_2, $\mu(t_2)$, maximum queue length $Q(t_2)$, density at t_2, $k(t_2)$, backward wave speed w_b, total delay W, average delay $\bar{w}$
Step 2: Reduced net flow calculation: The adjusted net flow π' is calculated by multiplying the original flow π by a reduction factor, which is expressed as a percentage. (i) **Demand-side strategy:** The updated arrival rate is $\lambda'(t) = \lambda(t) - (\pi - \pi')$, while the flow throughput remains unchanged $\mu'(t) = \mu(t)$; (ii) **Supply-side strategy:** The new flow throughput is given by $\mu'(t) = Cap - \pi' \cdot \theta \cdot (t - t_0)$, with the arrival rate $\lambda'(t) = \mu'(t) + \pi'$; (iii) **Demand-supply coupled strategy:** The flow throughput is updated as $\mu'(t) = Cap - \pi' \cdot \theta \cdot (t - t_0)$, while ensuring equilibrium between arrival and flow throughput $\lambda'(t) = \mu'(t) + \pi'$.
Step 3: Arrival and departure rate calculation: Calculated the updated arrival and flow throughput at t_2, $\mu'(t_2) = Cap - \pi' \cdot \theta \cdot (t_2 - t_0)$, $\lambda'(t_2) = \mu'(t_2) + \pi'$.

Step 4: Performance parameters in FDQ-PW calculation:

(i) The maximum queue length is estimated by $Q'(t_2) = \pi' \cdot (t_2 - t_0)$

(ii) The total delay is estimated by $W' \approx \Delta t \cdot \sum_{t_0}^{t_3} Q'(t)$

(iii) The average delay is estimated by $\bar{w}' = W' / \sum_{t_0}^{t_3} Q'(t)$

Step 5: Density calculation:

The updated traffic density at t_2 is $k'(t_2) = \frac{Q'(t_2)}{\varphi} + k_c$, where φ represents the spatial extent of queued vehicles.

Step 6: Parameters in FD calculation:

(i) The changed backward wave speed is estimated by ${w_b}' = \frac{Cap - \mu'(t_2)}{k'(t_2) - k_c}$,

(ii) The changed jam density is estimated by ${k_j}' = \frac{Cap - \mu'(t_2)}{k'(t_2) - k_c}$

Step 7: Efficiency loss and gain analysis:

(i) Efficiency gain in maximum queue length: Compare the maximum queue length before and after implementing the strategy and the percentage reduction in the maximum queue length is given by:

$$\Gamma_Q = \frac{Q(t_2) - Q'(t_2)}{Q(t_2)} \times 100\%$$

(ii) Efficiency gain in total delay: Compare the total delay before and after implementing the strategy and the relative reduction in total delay is:

$$\Gamma_W = \frac{W - W'}{W} \times 100\%$$

(iii) Efficiency gain in average delay: Compare the average delay before and after implementing the strategy and the relative reduction in average delay is:

$$\Gamma_{\bar{w}} = \frac{\bar{w} - \bar{w}'}{\bar{w}} \times 100\%$$

Statement of Contribution

Yuyan (Annie) Pan: Formal analysis, Validation, Visualization, Writing-original draft, Writing-review and editing. **Xianbiao Hu**: Methodology, Supervision, Resources, Funding acquisition, Writing-original draft, Writing-review and editing. **Xuesong (Simon) Zhou**: Conceptualization, Methodology, Writing-original draft, Writing-review and editing.

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No generative AI tools were used to create scientific content, formulate methods, derive equations, analyze data, or develop conceptual or technical ideas in this manuscript. All modeling, derivations, interpretations, and conclusions were produced entirely by the authors. Generative AI tools (e.g., ChatGPT) were used only for language proofreading and minor editorial improvements, and the authors take full responsibility for the final content.

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